
EECS 16A Designing Information Devices and Systems I

Discussion 0B

1. Solving Systems of Equations A system of linear equations can either have one solution, an infinite number of solutions, or no solution at all. For the following systems of equations, state whether there is a unique solution, no solution, or an infinite number of solutions.

(a) Determine the values of x and y that satisfy the following system of equations:

$$2x + 3y = 7$$

$$4x - y = 5$$

(b) Determine the values of x and y that satisfy the following system of equations:

$$2x + 5y = 12$$

$$4x + 10y = 8$$

(c) Determine the values of x , y , and z that satisfy the following system of equations:

$$2x + y - z = 3$$

$$4x + 2y - 2z = 6$$

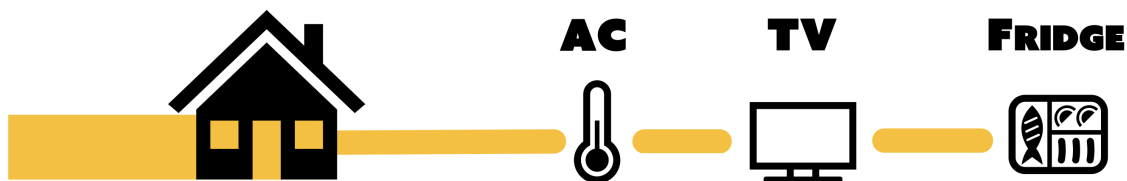
$$3x - y + 2z = 4$$

EECS 16A Designing Information Devices and Systems I

Discussion 1A

1. Energy Disaggregation

Recently, energy companies (like PG&E) have invested heavily into a problem called *energy disaggregation*. This is where you take measurements of the total electricity drawn from a home and then try to deduce which appliances are running in that house. Energy companies value this information since it helps them predict the amount of electricity they will need to produce on a given day, which can be particularly useful during heat waves or power cuts due to wildfire danger. They can also offer their customers energy-saving recommendations.



Let's try solving the *energy disaggregation* problem ourselves!

Suppose you live in a home with just these **3 appliances; an air conditioning unit (AC), a television (TV), and a refrigerator (R)**. Now say you want to find the amount of electricity these appliances use individually, but the only measurement you can take is of the total power your home draws using your meter outside (this is often mounted on the side of the house and shows a running total of your electricity usage).

To do this you will turn some appliances on and off and then read different total measurements. You can turn off the TV at any time, but you **can't unplug the fridge** since the food would spoil. We keep the air conditioner off throughout the morning, but then it must stay on during the afternoon. However, the breaker trips (meaning the electricity suddenly shuts off) if all three are running, so **the TV and AC cannot run at the same time**.

- (a) Can you design a way to calculate how much power each appliance uses? What type of measurements will you need to make, and how many?

Let x_R be the power consumed by the refrigerator, x_{TV} by the TV and x_{AC} by the AC. To find out the values of three variables we somehow need three equations/measurements.

- (b) Write out a system of equations that describe your measurements. Can you solve this system so that each appliance's power is written in terms of measurements m_i ?

For example: If you measure the power m_1 in the afternoon with the AC and refrigerator on but the TV is off, then the equation might look like $x_{AC} + x_R = m_1$.

- (c) Let us say the breaker is fixed, so now we can safely run the AC and TV at the same time. Is there another way (or ways) you could create a new system of equations to solve? If so, see if you can solve your new system!

- (d) Lastly suppose, as a busy Berkeley student, you only get a chance to take two measurements. Can you determine how much power each of the three appliances draw? If not, what combinations of power consumption can you find out?

2. Solving Systems of Equations

While we'd love every system of linear equations to have a unique solution, in reality we can either have (a) one unique solution, (b) an infinite number of solutions, or (c) no solution at all. We are going to walk through some examples to see what sorts of equations create these types of solutions.

- (a) Let's consider the system (where each $a, b \in \mathbb{R}$ can be any real number):

$$\begin{aligned} ax + y &= 3 \\ -x + 2y &= b \end{aligned}$$

For each of the selected values for a and b , sketch or plot out the lines $y(x)$ each of these equations form.

Can you conclude which values result in a unique solution? Infinite solutions? No solutions?

- i. $a = 1$, $b = 0$
- ii. $a = 0$, $b = 2$
- iii. $a = -1/2$, $b = 6$
- iv. $a = -1/2$, $b = 4$

- (b) Now, assume we are using the tomography imaging technique described in lecture to image a 2x2 grid, as shown below.

2x2 Tomography Example

x_1	x_2
x_3	x_4

- i. We record the following measurements:

$$x_1 + x_2 = 2$$

$$x_1 + x_3 = 2$$

$$x_2 + x_4 = 2$$

$$x_3 + x_4 = 3$$

Solve this system using any method. Is this a valid set of measurements? Why or why not?

- ii. We are led to believe our measurements might be faulty, so we record the following new measurements:

$$x_1 + x_2 = 2$$

$$x_1 + x_3 = 3$$

$$x_2 + x_4 = 2$$

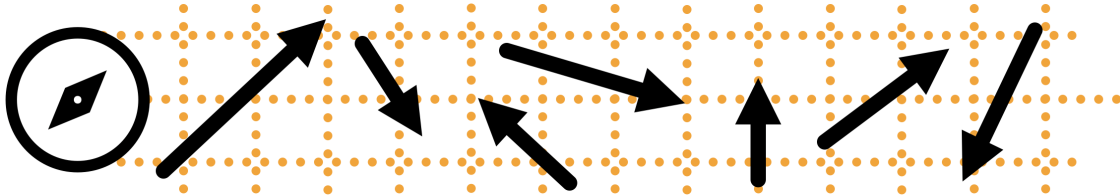
$$x_3 + x_4 = 3$$

Now, solve this system using any method. Is this a valid set of measurements? Why or why not?

EECS 16A Designing Information Devices and Systems I

Discussion 1B

1. Vectors



A vector is an ordered list of numbers. For instance, a point on a plane (x,y) is a vector! We label vectors using an arrow overhead $\vec{v}$, and since vectors can live in ANY dimension of space we'll need to leave our notation general $(x,y) \rightarrow \vec{v} = (v_1, v_2, \dots)$. Below are few more examples (the left-most form is the general definition):

$$\vec{x} = \begin{bmatrix} x_1 \\ \vdots \\ x_n \end{bmatrix} \in \mathbb{R}^n \qquad \vec{a} = \begin{bmatrix} 3 \\ 1 \\ 4 \end{bmatrix} \in \mathbb{R}^3 \qquad \vec{b} = \begin{bmatrix} 2.4 \\ 5.3 \end{bmatrix} \in \mathbb{R}^2$$

Just to unpack this a bit more, $\vec{b} \in \mathbb{R}^3$ in english means "vector $\vec{b}$ lives in 3-Dimensional space".

- The $\in$ symbol literally means "in"
- The $\mathbb{R}$ stands for "real numbers" (FUN FACT: $\mathbb{Z}$ means "integers" like $-2, 4, 0, \dots$)
- The exponent $\mathbb{R}^n \leftarrow$ indicates the dimension of space, or the amount of numbers in the vector.

One last thing: it is standard to write vectors in column-form, like seen with $\vec{a}, \vec{b}, \vec{x}$ above. We call these *column vectors*, in contrast to horizontally written vectors which we call *row vectors*.

Okay, let's dig into a few examples:

(a) Which of the following vectors live in $\mathbb{R}^2$ space?

$$i. \begin{bmatrix} 3 \\ 6 \end{bmatrix} \qquad ii. \begin{bmatrix} 5 \\ 0 \\ 3 \\ 5 \end{bmatrix} \qquad iii. \begin{bmatrix} -4.76 \\ 1.32 \\ 0.01 \end{bmatrix} \qquad iv. \begin{bmatrix} -20 \\ 100 \end{bmatrix}$$

(b) Graphically show the vectors (either in a sketch with axes, or a plot on a computer):

$$i. \begin{bmatrix} 2 \\ 5 \end{bmatrix}$$

$$ii. \begin{bmatrix} 5 \\ 2 \end{bmatrix}$$

(c) Compute the sum $\vec{a} + \vec{b} = \vec{c}$ from the vectors below, and then graphically sketch or plot these vectors. (show them in a way that forms a triangle; also is there only one possible triangle?)

$$\vec{a} = \begin{bmatrix} 1 \\ 2 \end{bmatrix}$$

$$\vec{b} = \begin{bmatrix} 4 \\ 3 \end{bmatrix}$$

2. Solving Systems of Equations A system of linear equations can either have one solution, an infinite number of solutions, or no solution at all. For the following systems of equations, state whether there is a unique solution, no solution, or an infinite number of solutions. If there are an infinite number of solutions give one possible solution.

(a) Solve the following system. How many solutions does it have?

$$\begin{array}{rcl} x & + & y = 4 \\ x & - & y = 2 \end{array}$$

(b) Now write the system in augmented matrix form:

$$\begin{array}{rcl} x + y & = & 4 & (1) \\ x - y & = & 2 & (2) \end{array}$$

- (c) Once in augmented matrix form we can use a systematic procedure called Gaussian Elimination to solve the system of equations. See what solution you get using Gaussian elimination.

- (d) Now consider the system

$$7x + y = 7 \quad (3)$$

$$42x + 6y = 42. \quad (4)$$

How many solutions does it have? Solve it first using any method, then write it as an augmented matrix and try to solve it.

(e) Now consider the system

$$7x + y = 7 \quad (5)$$

$$42x + 6y = 42 \quad (6)$$

$$7x + y = 6 \quad (7)$$

How many solutions does it have? Solve it first using any method, then write it as an augmented matrix and try to solve it.

EECS 16A Designing Information Devices and Systems I

Discussion 2A

1. Gaussian Elimination

Gaussian Elimination is a systematic procedure that a computer could follow for solving a large system of linear equations simultaneously. The augmented matrix is in the form:

$$\left[\begin{array}{cccc|c} a_{11} & a_{12} & \cdots & a_{1n} & b_1 \\ a_{21} & a_{22} & \cdots & a_{2n} & b_2 \\ \vdots & & & \vdots & \vdots \\ a_{m1} & a_{m2} & \cdots & a_{mn} & b_m \end{array} \right] \quad (1)$$

In Gaussian Elimination, we perform three operations on the system of equations. None of these three operations change the solution set of the system of equations. These operations are:

- (a) Multiplying an equation by a *nonzero* scalar constant.
- (b) Adding a scalar constant multiple of one equation to another.
- (c) Swapping two equations.

Gaussian Elimination Algorithm for augmented matrix (1):

Step 1: For $i = 1, 2, \dots, m$

- (i) If necessary, swap row i with a row below it, so that the leading entry in row i is as far left as possible.
- (ii) Rescale row i so that its leading entry is equal to 1.
- (iii) For rows $j = i + 1, i + 2, \dots, m$, add to row j a scalar multiple of row i , so that the leading entry of row i has all zeros below it.

Step 2: For $i = m, m - 1, \dots, 1$

- (i) Add to each row $j = 1, 2, \dots, i - 1$ a scalar multiple of row i so that the leading entry of row i has all zeros above it.

For the following systems, use Gaussian Elimination to solve the problem. Does a solution exist? Is it unique? If there are an infinite number of solutions, give the solution in the parametric form.

- (a) Let the three variables be x_1, x_2, x_3 . Solve the following augmented matrix form using Gaussian Elimination.

$$\left[\begin{array}{ccc|c} 3 & -1 & 2 & 1 \\ 0 & 0 & 2 & 1 \end{array} \right]$$

- (b) True or False: A system of equations with more equations than unknowns will always have either infinite solutions or no solutions.

(c)

$$\begin{bmatrix} 2 & 0 & 4 \\ 0 & 1 & 2 \\ 1 & 2 & 0 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} 6 \\ -3 \\ 3 \end{bmatrix}$$

(d)

$$\begin{bmatrix} 1 & 4 & 2 \\ 1 & 2 & 8 \\ 1 & 3 & 5 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} 2 \\ 0 \\ 3 \end{bmatrix}$$

(e)

$$\begin{bmatrix} 2 & 2 & 3 \\ 0 & 1 & 1 \\ 2 & 0 & 1 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} 7 \\ 3 \\ 1 \end{bmatrix}$$

EECS 16A Designing Information Devices and Systems I

Discussion 2B

1. Gaussian Elimination

Gaussian Elimination is a systematic procedure that a computer could follow for solving a large system of linear equations simultaneously. The augmented matrix is in the form:

$$\left[\begin{array}{cccc|c} a_{11} & a_{12} & \cdots & a_{1n} & b_1 \\ a_{21} & a_{22} & \cdots & a_{2n} & b_2 \\ \vdots & & & \vdots & \vdots \\ a_{m1} & a_{m2} & \cdots & a_{mn} & b_m \end{array} \right] \quad (1)$$

In Gaussian Elimination, we perform three operations on the system of equations. None of these three operations change the solution set of the system of equations. These operations are:

- (a) Multiplying an equation by a *nonzero* scalar constant.
- (b) Adding a scalar constant multiple of one equation to another.
- (c) Swapping two equations.

Gaussian Elimination Algorithm for augmented matrix (1):

Step 1: For $i = 1, 2, \dots, m$

- (i) If necessary, swap row i with a row below it, so that the leading entry in row i is as far left as possible.
- (ii) Rescale row i so that its leading entry is equal to 1.
- (iii) For rows $j = i + 1, i + 2, \dots, m$, add to row j a scalar multiple of row i , so that the leading entry of row i has all zeros below it.

Step 2: For $i = m, m - 1, \dots, 1$

- (i) Add to each row $j = 1, 2, \dots, i - 1$ a scalar multiple of row i so that the leading entry of row i has all zeros above it.

For the following systems, use Gaussian Elimination to solve the problem. Does a solution exist? Is it unique? If there are an infinite number of solutions, give the solution in the parametric form.

(a)

$$\begin{bmatrix} 2 & 0 & 4 \\ 0 & 1 & 2 \\ 1 & 2 & 0 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} 6 \\ -3 \\ 3 \end{bmatrix}$$

(b)

$$\begin{bmatrix} 1 & 4 & 2 \\ 1 & 2 & 8 \\ 1 & 3 & 5 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} 2 \\ 0 \\ 3 \end{bmatrix}$$

(c)

$$\begin{bmatrix} 2 & 2 & 3 \\ 0 & 1 & 1 \\ 2 & 0 & 1 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} 7 \\ 3 \\ 1 \end{bmatrix}$$

2. Finding The Bright Cave

Nara the one-handed druid and Kody the one-handed ranger find themselves in dire straits. Before them is a cliff with four cave entrances arranged in a square: two upper caves and two lower caves. Each entrance emits a certain amount of light, and the two wish to find exactly the amount of light coming from each cave. Here's the catch: after contracting a particularly potent strain of ghoul fever, our intrepid heroes are only able to see the total intensity of light before them (so their eyes operate like a single-pixel camera). Kody and Nara are capable adventurers, but they don't know any linear algebra – and they need your help.

Kody proposes an imaging strategy where he uses his hand to completely block the light from two caves at a time. He is able to take measurements using the following four masks (black means the light is blocked from that cave):

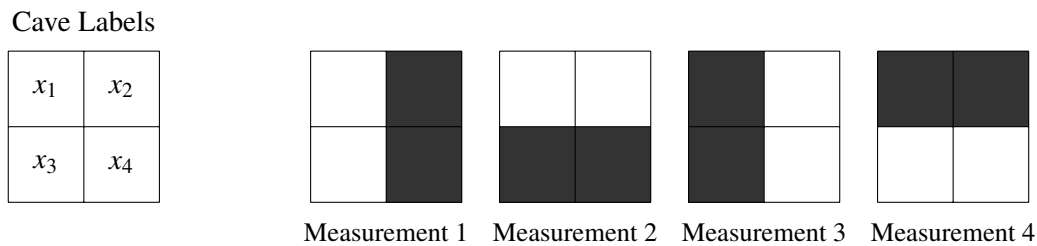


Figure 1: Four image masks.

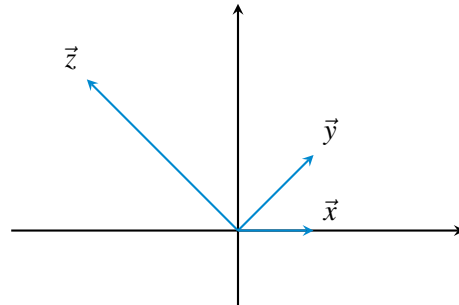
- (a) Let $\vec{x}$ be the four-element vector that represents the magnitude of light emanating from the four cave entrances. Write a matrix $\mathbf{K}$ that performs the masking process in Figure 1 on the vector $\vec{x}$, such that $\mathbf{K}\vec{x}$ is the result of the four measurements.
- (b) Does Kody's set of masks give us a unique solution for all four caves' light intensities? Why or why not?
- (c) Nara, in her infinite wisdom, places her one hand diagonally across the entrances, covering two of the cave entrances. However, her hand is not wide enough, letting in 50% of the light from the caves covered and 100% of the light from the caves not covered. The following diagram shows the percentage of light let through from each cave:

50%	100%
100%	50%

Does this additional measurement give them enough information to solve the problem? Why or why not?

3. Visualizing Span

We are given a point $\vec{z}$ that we want to get to, but we can only move in two directions: $\vec{x}$ and $\vec{y}$. We know that to get to $\vec{z}$, we can travel along $\vec{x}$ for some amount α , then change direction, and travel along $\vec{y}$ for some amount β . We want to find these two scalars α and β , such that we reach point $\vec{z}$. That is, $\alpha\vec{x} + \beta\vec{y} = \vec{z}$.



- (a) First, consider the case where $\vec{x} = \begin{bmatrix} 1 \\ 1 \end{bmatrix}$, $\vec{y} = \begin{bmatrix} 2 \\ 1 \end{bmatrix}$, and $\vec{z} = \begin{bmatrix} -2 \\ 2 \end{bmatrix}$. Draw these vectors on a sheet of paper.
- (b) We want to find the two scalars α and β , such that by moving α along $\vec{x}$ and β along $\vec{y}$ so that we can reach $\vec{z}$. Write a system of equations to find α and β in matrix form.
- (c) Solve for α, β .

EECS 16A Designing Information Devices and Systems I

Discussion 3A

Definition - Linear Dependence (I): A set of vectors $\{\vec{v}_1, \dots, \vec{v}_n\}$ is linearly dependent if there exist scalars $\alpha_1, \dots, \alpha_n$ such that $\alpha_1 \vec{v}_1 + \dots + \alpha_n \vec{v}_n = \vec{0}$ and not all α_i 's are equal to zero.

Definition - Linear Dependence (II): A set of vectors $\{\vec{v}_1, \dots, \vec{v}_n\}$ is linearly dependent if there exist scalars $\alpha_1, \dots, \alpha_n$ and an index i such that $\vec{v}_i = \sum_{j \neq i} \alpha_j \vec{v}_j$.¹ In words, a set of vectors is linearly dependent if one of the vectors could be written as a linear combination of the rest of the vectors.

These two definitions are equivalent. If we would like to prove linear dependence with definition (II), we need to first choose a vector $\vec{v}_i$ and show that it is a linear combination of the other vectors. However, with definition (I), we don't need to try to "single out" a vector to get started with the proof. We can blindly write down the equation $\alpha_1 \vec{v}_1 + \dots + \alpha_n \vec{v}_n = \vec{0}$ and begin our proof from there. On the other hand, definition (II) gives us a more intuitive way to talk about redundancy. If a vector can be constructed from the rest of the vectors, then this vector does not contribute any information that is not already captured by the other vectors.

Definition (Linear Independence): A set of vectors is linearly independent if it is not linearly dependent. More specifically, from the first definition of linear dependence we can deduce that a set of vectors $\{\vec{v}_1, \dots, \vec{v}_n\}$ is linearly independent if $\alpha_1 \vec{v}_1 + \dots + \alpha_n \vec{v}_n = \vec{0}$ implies $\alpha_1 = \dots = \alpha_n = 0$.

For more details, please refer to [Note 3](#).

Properties of Matrix-Vector Multiplication:

- (a) $\mathbf{A}(\vec{x} + \vec{y}) = \mathbf{A}\vec{x} + \mathbf{A}\vec{y}$ (Distributivity)
- (b) $\mathbf{A}\vec{x} \neq \vec{x}\mathbf{A}$
- (c) For $\mathbf{A}\vec{x}$ to be valid, if $A \in \mathbb{R}^{m \times n}$, then $\vec{x}$ must $\in \mathbb{R}^n$.
- (d)

$$\begin{bmatrix} | & | & | & | \\ \vec{a}_1 & \vec{a}_2 & \dots & \vec{a}_n \\ | & | & | & | \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ \vdots \\ x_n \end{bmatrix} = x_1 \vec{a}_1 + x_2 \vec{a}_2 + \dots + x_n \vec{a}_n = \sum_{i=1}^n x_i \vec{a}_i$$

¹In case you are unfamiliar with this notation, the $\sum$ symbol is simply shorthand for addition. For instance, $\alpha_1 \vec{v}_1 + \dots + \alpha_n \vec{v}_n$ can be written as $\sum_{i=1}^n \alpha_i \vec{v}_i$ or $\sum_i \alpha_i \vec{v}_i$, which is a sum over all possible i values. In this instance, $\sum_{j \neq i} \alpha_j \vec{v}_j$ is the sum over all $\alpha_j \vec{v}_j$ excluding the $\alpha_i \vec{v}_i$ term, which can also be calculated as $\alpha_1 \vec{v}_1 + \dots + \alpha_n \vec{v}_n - \alpha_i \vec{v}_i$.

1. Proofs

- (a) Suppose for some non-zero vector $\vec{x}$, $\mathbf{A}\vec{x} = \vec{0}$. Prove that the columns of $\mathbf{A}$ are linearly dependent.
- (b) Suppose there exists a matrix $\mathbf{A}$ whose columns are linearly dependent. Prove that if there exists a solution to $\mathbf{A}\vec{x} = \vec{b}$, then there are infinitely many solutions.
- (c) Suppose there exist two unique vectors $\vec{x}_1$ and $\vec{x}_2$ that both satisfy $\mathbf{A}\vec{x} = \vec{b}$, that is, $\mathbf{A}\vec{x}_1 = \vec{b}$ and $\mathbf{A}\vec{x}_2 = \vec{b}$. Prove that the columns of $\mathbf{A}$ are linearly dependent.

2. Span Proof

Given some set of vectors $\{\vec{v}_1, \vec{v}_2, \dots, \vec{v}_n\}$, show the following:

$$\text{span}\{\vec{v}_1, \vec{v}_2, \dots, \vec{v}_n\} = \text{span}\{\alpha \vec{v}_1, \vec{v}_2, \dots, \vec{v}_n\}, \text{ where } \alpha \text{ is a non-zero scalar.}$$

In other words, we can scale our spanning vectors and not change their span.

EECS 16A Designing Information Devices and Systems I

Discussion 3B

1. Matrix Multiplication

Consider the following matrices:

$$\mathbf{A} = \begin{bmatrix} 1 & 4 \end{bmatrix} \quad \mathbf{B} = \begin{bmatrix} 3 \\ 2 \end{bmatrix} \quad \mathbf{C} = \begin{bmatrix} 1 & 4 \\ 2 & 3 \end{bmatrix} \quad \mathbf{D} = \begin{bmatrix} 3 & 2 \\ 2 & 1 \end{bmatrix}$$
$$\mathbf{E} = \begin{bmatrix} 1 & 9 & 5 & 7 \\ 4 & 3 & 2 & 2 \end{bmatrix} \quad \mathbf{F} = \begin{bmatrix} 5 & 5 & 8 \\ 6 & 1 & 2 \\ 4 & 1 & 7 \\ 3 & 2 & 2 \end{bmatrix} \quad \mathbf{G} = \begin{bmatrix} 8 & 1 & 6 \\ 3 & 5 & 7 \\ 4 & 9 & 2 \end{bmatrix} \quad \mathbf{H} = \begin{bmatrix} 5 & 3 & 4 \\ 1 & 8 & 2 \\ 2 & 3 & 5 \end{bmatrix}$$

For each matrix multiplication problem, *if the product exists*, find the product by hand. Otherwise, explain why the product does not exist.

(a) $\mathbf{A B}$

(b) $\mathbf{C D}$

(c) $\mathbf{D C}$

(d) $\mathbf{C E}$

(e) **F E**

(f) **E F**

(g) **G H** (Practice on your own)

(h) **H G** (Practice on your own)

2. Visualizing Matrices as Operations

This problem is going to help you visualize matrices as operations. For example, when we multiply a vector by a “rotation matrix,” we will see it “rotate” in the true sense here. Similarly, when we multiply a vector by a “reflection matrix,” we will see it be “reflected.” The way we will see this is by applying the operation to all the vertices of a polygon and seeing how the polygon changes.

Your TA will now show you how a unit square can be rotated, scaled, or reflected using matrices!

Part 1: Rotation Matrices as Rotations

- (a) We are given matrices $\mathbf{T}_1$ and $\mathbf{T}_2$, and we are told that they will rotate the unit square by 15° and 30° , respectively. Design a procedure to rotate the unit square by 45° using only $\mathbf{T}_1$ and $\mathbf{T}_2$, and plot the result in the IPython notebook. How would you rotate the square by 60° ?

(b) Try to rotate the unit square by 60° using only one matrix. What does this matrix look like?

(c) $\mathbf{T}_1$, $\mathbf{T}_2$, and the matrix you used in part (b) are called “rotation matrices.” They rotate any vector by an angle θ . Show that a rotation matrix has the following form:

$$\mathbf{R} = \begin{bmatrix} \cos \theta & -\sin \theta \\ \sin \theta & \cos \theta \end{bmatrix}$$

where θ is the angle of rotation. To do this consider rotating the unit vector $\begin{bmatrix} \cos(\alpha) \\ \sin(\alpha) \end{bmatrix}$ by θ degrees using the matrix $\mathbf{R}$.

(Definition: A vector, $\vec{v} = \begin{bmatrix} v_1 \\ v_2 \\ \vdots \end{bmatrix}$, is a unit vector if $\sqrt{v_1^2 + v_2^2 + \dots} = 1$.)

(Hint: Use your trigonometric angle sum identities: $\cos(\theta_1 + \theta_2) = \cos(\theta_1)\cos(\theta_2) - \sin(\theta_1)\sin(\theta_2)$ and $\sin(\theta_1 + \theta_2) = \sin(\theta_1)\cos(\theta_2) + \cos(\theta_1)\sin(\theta_2)$)

(d) **(Practice)** Now, we want to get back the original unit square from the rotated square in part (b). What matrix should we use to do this? **(Note:** Don’t use inverses! Answer this question using your intuition, we will visit inverses very soon in lecture!)

(e) **(Practice)** Use part (d) to obtain the “inverse” rotation matrix for a matrix that rotates a vector by θ . Multiply the inverse rotation matrix with the rotation matrix and vice-versa. What do you get?

- (f) **(Practice)** What are the matrices that reflect a vector about the (i) x -axis, (ii) y -axis, and (iii) $x = y$

Part 2: Commutativity of Operations

A natural question to ask is the following: Does the *order* in which you apply these operations matter? Follow your TA to obtain the answers to the following questions!

- (a) Let's see what happens to the unit square when we rotate the square by 60° and then reflect it along the y -axis.
- (b) Now, let's see what happens to the unit square when we first reflect the square along the y -axis and then rotate it by 60° .
- (c) Try to do steps (a) and (b) by multiplying the reflection and rotation matrices together (in the correct order for each case). What does this tell you?
- (d) If you reflected the unit square twice (along any pair of axes), do you think the order in which you applied the reflections would matter? Why/why not?

(Practice) Part 3: Distributivity of Operations

- (a) The distributivity property of matrix-vector multiplication holds for any vectors and matrices. Show for general $\mathbf{A} \in \mathbb{R}^{2 \times 2}$ and $\vec{v}_1, \vec{v}_2 \in \mathbb{R}^2$ that $\mathbf{A}(\vec{v}_1 + \vec{v}_2) = \mathbf{A}\vec{v}_1 + \mathbf{A}\vec{v}_2$.

EECS 16A Designing Information Devices and Systems I

Discussion 4A

1. Mechanical Inverses

A square matrix A is said to be invertible if there exists a matrix B such that $AB = I$ and $BA = I$. The matrix B is called the inverse of A and is denoted by A^{-1} .

Some properties of the inverse are:

- $(A^{-1})^{-1} = A$
- $(A^T)^{-1} = (A^{-1})^T$

In each part, determine whether the inverse of $\mathbf{A}$ exists. If it exists, find it.

(a) $\mathbf{A} = \begin{bmatrix} 1 & 0 \\ 0 & 9 \end{bmatrix}$

(b) $\mathbf{A} = \begin{bmatrix} a & b \\ c & d \end{bmatrix}$

(c) (PRACTICE)

$$\mathbf{A} = \begin{bmatrix} 5 & 4 \\ 1 & 1 \end{bmatrix}$$

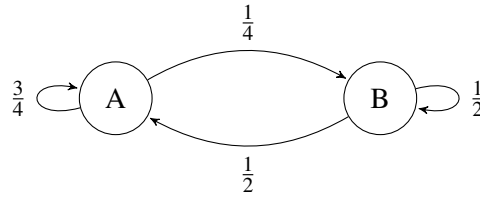
(d) $\mathbf{A} = \begin{bmatrix} 5 & 5 & 15 \\ 2 & 2 & 4 \\ 1 & 1 & 4 \end{bmatrix}$

(e) (PRACTICE)

$$\mathbf{A} = \begin{bmatrix} 5 & 5 & 15 \\ 2 & 2 & 4 \\ 1 & 0 & 4 \end{bmatrix}$$

2. Transition Matrix

Suppose we have a network of pumps as shown in the diagram below. Let us describe the state of A and B using a state vector $\vec{x}[n] = \begin{bmatrix} x_A[n] \\ x_B[n] \end{bmatrix}$ where $x_A[n]$ and $x_B[n]$ are the states at time-step n .



- Find the state transition matrix S , such that $\vec{x}[n+1] = S \vec{x}[n]$.
Separately find the sum of the terms for each column vector in S . Do you notice any pattern?
- Let us now find the matrix S^{-1} such that we can recover the previous state $\vec{x}[n-1]$ from $\vec{x}[n]$.
Specifically, solve for S^{-1} such that $\vec{x}[n-1] = S^{-1} \vec{x}[n]$.
- Now draw the state transition diagram that corresponds to the S^{-1} that you just found.
Also find the sum of the terms for each column vector in S^{-1} . Do you notice any pattern?
- Redraw the diagram from the first part of the problem, but now with the directions of the arrows reversed. Let us call the state transmission matrix of this "reversed" state transition diagram T . Does $T = S^{-1}$?
- Suppose we start in the state $\vec{x}[1] = \begin{bmatrix} 12 \\ 12 \end{bmatrix}$. Compute the state vector after 2 time-steps $\vec{x}[3]$.
- (Challenge practice problem)** Given our starting state from the previous problem, what happens if we look at the state of the network after a lot of time steps? Specifically which state are we approaching, as defined below?

$$\vec{x}_{final} = \lim_{n \rightarrow \infty} \vec{x}[n]$$

Note that the final state needs to be what we call a *steady state*, meaning $S \vec{x}_{final} = \vec{x}_{final}$.

Also what can you say about $x_A[n] + x_B[n]$?

Use information from both of these properties to write out a new system of equations and solve for $\vec{x}_{final}$.

EECS 16A Designing Information Devices and Systems I

Discussion 4B

1. Column Spaces and Null Spaces Intro

- The **column space** is the possible outputs of a transformation/function/linear operation. It is also the **span** of the column vectors of the matrix.
- The **null space** is the set of input vectors that output the zero vector.

For the following matrices, answer the following questions:

- What is the column space of $\mathbf{A}$? What is its dimension?
- What is the null space of $\mathbf{A}$? What is its dimension?
- Are the column spaces of the row reduced matrix $\mathbf{A}$ and the original matrix $\mathbf{A}$ the same?
- Do the columns of $\mathbf{A}$ form a basis for $\mathbb{R}^2$? Why or why not?

(a) $\begin{bmatrix} 0 & 1 \\ 0 & 1 \end{bmatrix}$

(b) $\begin{bmatrix} 1 & 2 \\ -1 & 1 \end{bmatrix}$

(c) $\begin{bmatrix} -2 & 4 \\ 3 & -6 \end{bmatrix}$

2. Identifying a Basis

Does each of these sets of vectors describe a basis for $\mathbb{R}^3$? If the vectors do not form a basis for $\mathbb{R}^3$, can they be thought of as a basis for some other vector space? If so, write an expression describing this vector space.

$$V_1 = \left\{ \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix}, \begin{bmatrix} 1 \\ 0 \\ 1 \end{bmatrix} \right\} \quad V_2 = \left\{ \begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix}, \begin{bmatrix} 1 \\ 0 \\ 1 \end{bmatrix}, \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix} \right\} \quad V_3 = \left\{ \begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix}, \begin{bmatrix} 1 \\ 0 \\ 1 \end{bmatrix}, \begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix} \right\}$$

3. Identifying a Subspace: Proof

Is the set

$$V = \left\{ \vec{v} \mid \vec{v} = c \begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix} + d \begin{bmatrix} 1 \\ 0 \\ 1 \end{bmatrix}, \text{ where } c, d \in \mathbb{R} \right\}$$

a subspace of $\mathbb{R}^3$? Why/why not?

EECS 16A Designing Information Devices and Systems I

Discussion 5A

1. Proof

Concept: Null Spaces, Invertibility

Consider a square matrix $\mathbf{A}$. Prove that if $\mathbf{A}$ has a non-trivial nullspace, i.e. if the nullspace of $\mathbf{A}$ contains more than just $\vec{0}$, then the matrix $\mathbf{A}$ is not invertible.

2. Steady State Reservoir Levels

We have 3 reservoirs: A, B and C . The pumps system between the reservoirs is depicted in the figure below.

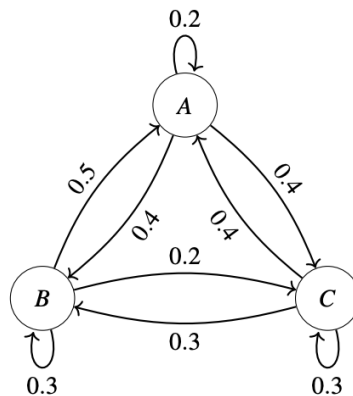


Figure 1: Reservoir pumps system.

(a) Write out the transition matrix $\mathbf{T}$ representing the pumps system.

- (b) You are told that $\lambda_1 = 1$, $\lambda_2 = \frac{-\sqrt{2}-1}{10}$, $\lambda_3 = \frac{\sqrt{2}-1}{10}$ are the eigenvalues of $\mathbf{T}$. Find a steady state vector $\vec{x}$, i.e. a nonzero vector such that $T\vec{x} = \vec{x}$.

3. Row Operations and Determinants

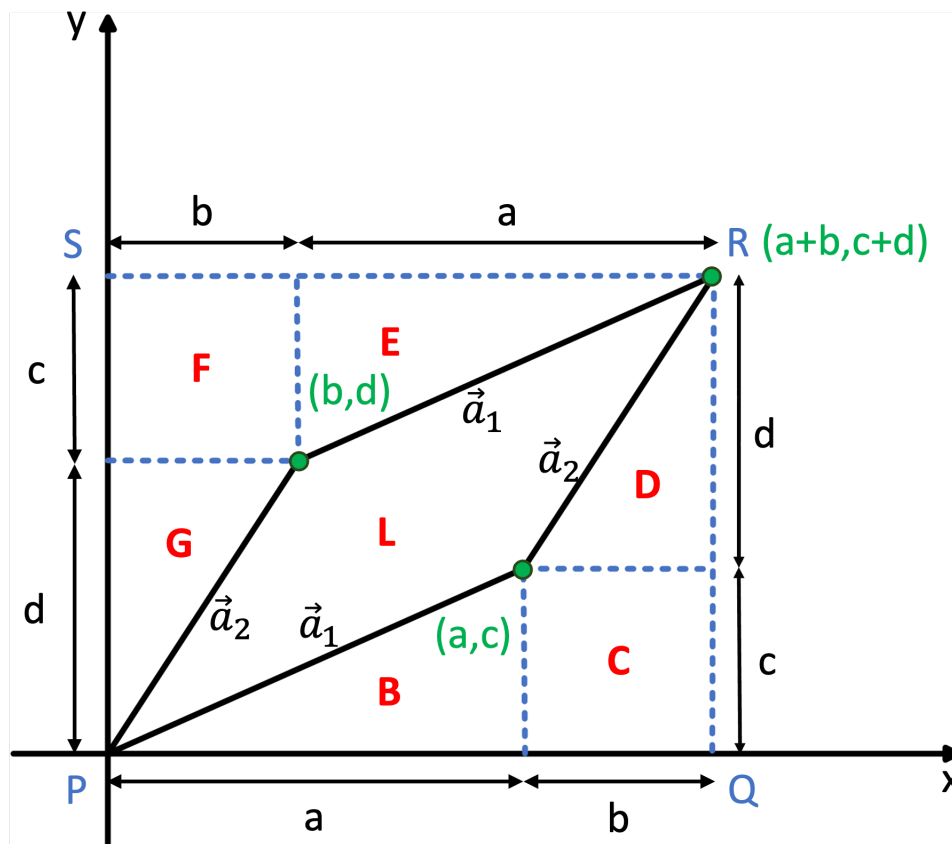


Figure 1: Parallelogram formed by columns of $\mathbf{A}$

Recall from lecture the way to compute a determinant of any 2×2 matrix is by using the following formula:

$$\mathbf{A} = \begin{bmatrix} a & b \\ c & d \end{bmatrix} \quad \det(\mathbf{A}) = ad - bc$$

There is a geometric interpretation of the determinant of a 2×2 matrix. The determinant of a 2×2 matrix $\mathbf{A}$ is the area of the parallelogram formed by the columns of $\mathbf{A}$. This generalizes to higher dimensions as well. For example, the determinant of a 3×3 matrix is the volume of the parallelepiped formed by the columns of the 3×3 matrix. In general, the determinant of an $n \times n$ matrix is the n -dimensional volume formed by the columns of that matrix.

We will show that this geometric relationship holds in the case of a 2×2 matrix. Consider the following matrix:

$$\mathbf{A} = \begin{bmatrix} a & b \\ c & d \end{bmatrix}$$

This matrix has columns:

$$\vec{a}_1 = \begin{bmatrix} a \\ c \end{bmatrix}$$

$$\vec{a}_2 = \begin{bmatrix} b \\ d \end{bmatrix}$$

Figure 1 shows the parallelogram formed by the columns of $\mathbf{A}$.

To find the area of this parallelogram, we can find the area of the rectangle PQRS and subtract from this the areas of the triangles and smaller rectangles that are not part of the parallelogram. We can see from the figure that the area of rectangle PQRS is $(a+b)(c+d)$. Expanding, we get $\text{area}(\text{PQRS}) = ac + ad + bc + bd$. We'll call this area H , and we want to find the area denoted by L in the figure. We can see that $L = H - B - C - D - E - F - G$. Also, we can use geometry to find each area in terms of the entries of $\mathbf{A}$.

$$H = ac + ad + bc + bd$$

$$B = \frac{1}{2}ac$$

$$C = bc$$

$$D = \frac{1}{2}bd$$

$$E = \frac{1}{2}ac$$

$$F = bc$$

$$G = \frac{1}{2}bd$$

Substituting these into $L = H - B - C - D - E - F - G$, we get:

$$L = ac + ad + bc + bd - \frac{1}{2}ac - bc - \frac{1}{2}bd - \frac{1}{2}ac - bc - \frac{1}{2}bd = ad - bc$$

So, we have that the area of the parallelogram formed by the columns of $\mathbf{A}$ is $ad - bc = \det(\mathbf{A})$.

Compute the determinant of the following matrix using the formula above:

(a) $\begin{bmatrix} 1 & 1 \\ 2 & 3 \end{bmatrix}$

EECS 16A Designing Information Devices and Systems I

Discussion 5B

1. Mechanical Eigenvalues and Eigenvectors

In each part, find the eigenvalues of the matrix and the associated eigenspaces. State if the inverse of the matrix exists.

(a) $\mathbf{A} = \begin{bmatrix} 0 & 1 \\ -2 & -3 \end{bmatrix}$

(b) $\mathbf{A} = \begin{bmatrix} -2 & 4 \\ -4 & 8 \end{bmatrix}$

(c) $\mathbf{A} = \begin{bmatrix} 0 & 1 \\ 0 & 0 \end{bmatrix}$

(d) $\mathbf{M} = \begin{bmatrix} 0 & -1 \\ 1 & 0 \end{bmatrix}.$

(e) **(PRACTICE)** $\mathbf{M} = \begin{bmatrix} 1 & 0 \\ 0 & 9 \end{bmatrix}$

2. Steady and Unsteady States

(a) You're given the matrix $\mathbf{M}$:

$$\mathbf{M} = \begin{bmatrix} \frac{1}{2} & \frac{1}{2} & -\frac{1}{2} \\ 0 & 1 & -2 \\ 0 & 0 & 2 \end{bmatrix}$$

Which generates the next state of a physical system from its previous state: $\vec{x}[k+1] = \mathbf{M}\vec{x}[k]$. ($\vec{x}$ could describe people or water, for example) Find the eigenspaces associated with the following eigenvalues:

- i. $\text{span}(\vec{v}_1)$, associated with $\lambda_1 = 1$
- ii. $\text{span}(\vec{v}_2)$, associated with $\lambda_2 = 2$
- iii. $\text{span}(\vec{v}_3)$, associated with $\lambda_3 = \frac{1}{2}$

(b) Define $\vec{x} = \alpha\vec{v}_1 + \beta\vec{v}_2 + \gamma\vec{v}_3$, a linear combination of the eigenvectors. For each of the cases in the table, determine if

$$\lim_{n \rightarrow \infty} \mathbf{M}^n \vec{x}$$

converges. If it does, what does it converge to?

α	β	γ	Converges?	$\lim_{n \rightarrow \infty} \mathbf{M}^n \vec{x}$
0	0	$\neq 0$		
0	$\neq 0$	0		
0	$\neq 0$	$\neq 0$		
$\neq 0$	0	0		
$\neq 0$	0	$\neq 0$		
$\neq 0$	$\neq 0$	0		
$\neq 0$	$\neq 0$	$\neq 0$		

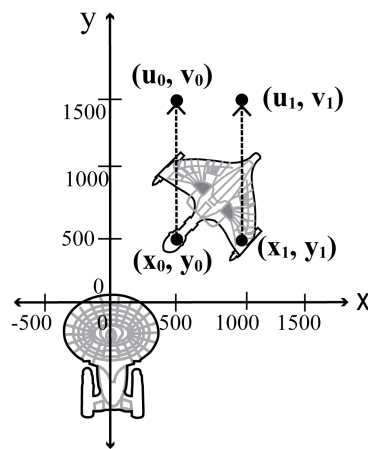
EECS 16A Designing Information Devices and Systems I

Discussion 6A

- 1. The Romulan Ruse** While scanning parts of the galaxy for alien civilization, the starship USS Enterprise NC-1701D encounters a Romulan starship that is known for advanced cloaking devices.

(a) *Concept: Matrix Transformations*

The Romulan illusion technology causes a point (x_0, y_0) to transform or *map* to (u_0, v_0) . Similarly, (x_1, y_1) is mapped to (u_1, v_1) . Figure 1 and Table 1 show two points on a Romulan ship and the corresponding *mapped* points.



Original Point	Mapped Point
$(x_0, y_0) = (500, 500)$	$(u_0, v_0) = (500, 1500)$
Original Point	Mapped Point
$(x_1, y_1) = (1000, 500)$	$(u_1, v_1) = (1000, 1500)$

Table 1: Original and Mapped Points

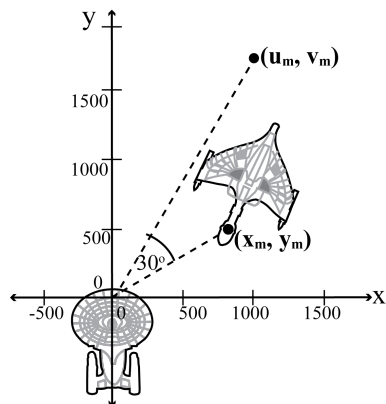
Figure 1: Figure for part (a)

Find a transformation matrix $\mathbf{A}_0$ such that

$$\begin{bmatrix} u_0 \\ v_0 \end{bmatrix} = \mathbf{A}_0 \begin{bmatrix} x_0 \\ y_0 \end{bmatrix}, \text{ and } \begin{bmatrix} u_1 \\ v_1 \end{bmatrix} = \mathbf{A}_0 \begin{bmatrix} x_1 \\ y_1 \end{bmatrix}.$$

(b) *Concept: Matrix Transformations*

In this scenario, every point on the Romulan ship (x_m, y_m) is mapped to (u_m, v_m) , such that vector $\begin{bmatrix} x_m \\ y_m \end{bmatrix}$ is rotated counterclockwise by 30° and then scaled by 2 in the x- and y-directions. This transformation is shown in Figure 2.



θ	$\sin \theta$	$\cos \theta$	$\tan \theta$
0°	0	1	0
30°	$\frac{1}{2}$	$\frac{\sqrt{3}}{2}$	$\frac{1}{\sqrt{3}}$
45°	$\frac{1}{\sqrt{2}}$	$\frac{1}{\sqrt{2}}$	1
60°	$\frac{\sqrt{3}}{2}$	$\frac{1}{2}$	$\sqrt{3}$
90°	1	0	∞

Table 2: Trigononometric Table

Figure 2: Figure for part (b)

Find a transformation matrix $\mathbf{R}$ such that $\begin{bmatrix} u_m \\ v_m \end{bmatrix} = \mathbf{R} \begin{bmatrix} x_m \\ y_m \end{bmatrix}$.

The Romulan ship has launched a probe into space and the Enterprise is trying to destroy the probe by firing a photon torpedo along a straight line from point $(0,0)$ towards the probe.

(c) *Concept: Gaussian Elimination, Systems of Equations*

The Romulan generals found a clever way to hide the probe by transforming (mapping) its position with a *cloaking* (transformation) matrix $\mathbf{A}_p$:

$$\mathbf{A}_p = \begin{bmatrix} 1 & 3 \\ 2 & 6 \end{bmatrix}.$$

They positioned the probe at (x_p, y_p) so that it maps to

$$(u_p, v_p) = (0, 0), \text{ where } \begin{bmatrix} u_p \\ v_p \end{bmatrix} = \mathbf{A}_p \begin{bmatrix} x_p \\ y_p \end{bmatrix}.$$

This scenario is shown in Figure 3. The initial position of the torpedo is $(0,0)$ and the torpedo cannot be fired on its initial position! Impressive trick indeed!

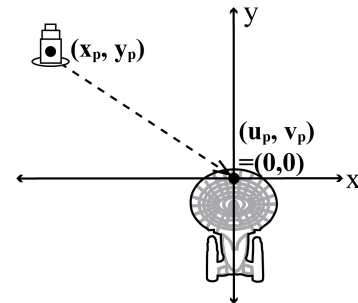


Figure 3: Figure for part (c)

Find the possible positions of the probe (x_p, y_p) so that $(u_p, v_p) = (0, 0)$.

(d) *Concept: Eigenspaces/Eigenvectors/Eigenvalues*

It turns out the Romulan engineers were not as smart as the Enterprise engineers. Their calculations did not work out and they positioned the probe at (x_q, y_q) such that the *cloaking* (transformation) matrix, $\mathbf{A}_p$, mapped it to $(u_q, v_q) \neq (0, 0)$, where

$$\begin{bmatrix} u_q \\ v_q \end{bmatrix} = \mathbf{A}_p \begin{bmatrix} x_q \\ y_q \end{bmatrix}, \text{ and } \mathbf{A}_p = \begin{bmatrix} 1 & 3 \\ 2 & 6 \end{bmatrix}.$$

As a result, the torpedo while traveling along a straight line from $(0, 0)$ to (u_q, v_q) , hit the probe at (x_q, y_q) on the way!

The scenario is shown in Figure 4. For the torpedo to hit the probe, we must have $\begin{bmatrix} u_q \\ v_q \end{bmatrix} = \lambda \begin{bmatrix} x_q \\ y_q \end{bmatrix}$, where λ is a real number.

Find the possible positions of the probe (x_q, y_q) so that $(u_q, v_q) = (\lambda x_q, \lambda y_q)$. Remember that the torpedo cannot be fired on its initial position $(0, 0)$.

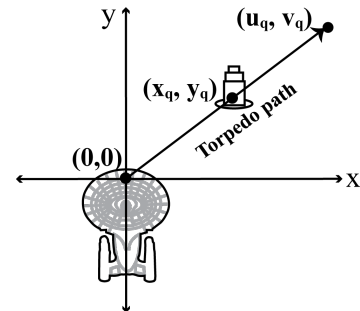


Figure 4: Figure for part (d)

EECS 16A Designing Information Devices and Systems I

Discussion 6B

1. Eigenvalues and Special Matrices – Visualization

As seen earlier, an eigenvector $\vec{v}$ belonging to a square matrix $\mathbf{A}$ is a nonzero vector that satisfies

$$\mathbf{A}\vec{v} = \lambda\vec{v}$$

where λ is a scalar known as the **eigenvalue** corresponding to eigenvector $\vec{v}$. Rather than mechanically computing the eigenvalues and eigenvectors, answer each part here by reasoning about the matrix at hand.

- (a) Does the identity matrix in $\mathbb{R}^n$ have any eigenvalues $\lambda \in \mathbb{R}$? What are the corresponding eigenvectors?

- (b) Does a diagonal matrix $\begin{bmatrix} d_1 & 0 & 0 & \cdots & 0 \\ 0 & d_2 & 0 & \cdots & 0 \\ 0 & 0 & d_3 & \cdots & 0 \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & 0 & \cdots & d_n \end{bmatrix}$ in $\mathbb{R}^n$ have any eigenvalues $\lambda \in \mathbb{R}$? What are the corresponding eigenvectors?

- (c) Conceptually, does $\mathbf{R}$, a rotation matrix in $\mathbb{R}^2$ by angle θ , have any eigenvalues $\lambda \in \mathbb{R}$? For which angles is this the case?

- (d) Now let us mechanically compute the eigenvalues of the rotation matrix in $\mathbb{R}^2$. Does it agree with our findings above? As a refresher, the rotation matrix $\mathbf{R}$ has the following form:

$$\mathbf{R} = \begin{bmatrix} \cos(\theta) & -\sin(\theta) \\ \sin(\theta) & \cos(\theta) \end{bmatrix}$$

- (e) Does the reflection matrix $\mathbf{T}$ across the x-axis in $\mathbb{R}^{2 \times 2}$ have any eigenvalues $\lambda \in \mathbb{R}$?

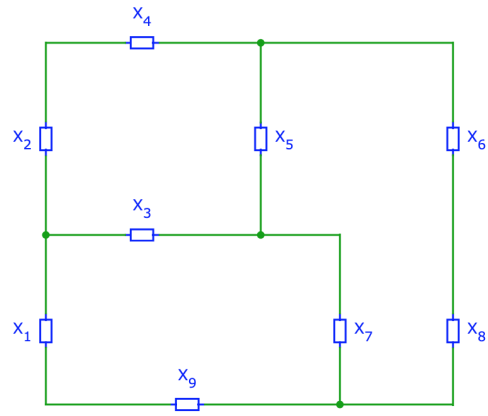
$$\mathbf{T} = \begin{bmatrix} 1 & 0 \\ 0 & -1 \end{bmatrix}$$

- (f) If a matrix $\mathbf{M}$ has an eigenvalue $\lambda = 0$, what does this say about its null space? What does this say about the solutions of the system of linear equations $\mathbf{M}\vec{x} = \vec{b}$?

- (g) **(Practice)** Does the matrix $\begin{bmatrix} 1 & 1 \\ 0 & 0 \end{bmatrix}$ have any eigenvalues $\lambda \in \mathbb{R}$? What are the corresponding eigenvectors?

2. Nodes

In the circuit shown below, label and count all nodes.

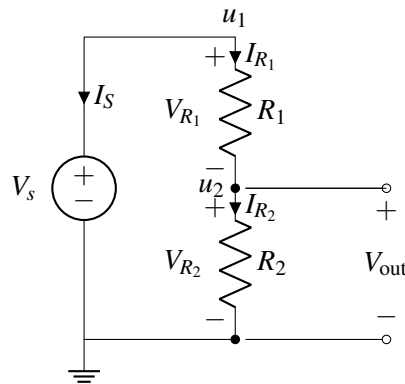


EECS 16A Designing Information Devices and Systems I

Discussion 7A

1. Voltage Divider

For the circuit below, your goal will be to find the voltage V_{out} in terms of the resistances R_1 , R_2 , and V_s , using NVA (Node Voltage Analysis). The labeling steps (steps 1-4) have already been done for you.

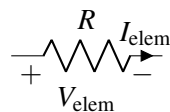


Here is a reminder of the labeling steps followed to get the circuit diagram above:

- **Step 1:** Select a reference (ground) node. Any node can be chosen for this purpose. We will measure all of the voltages in the rest of the circuit relative to this point.
- **Step 2:** Label all nodes with voltage set by voltage sources.
- **Step 3:** Label remaining nodes.
- **Step 4:** Label element voltages and currents, following **Passive Sign Convention** (discussed below).

Passive sign convention

The **passive sign convention** dictates that positive current should *enter* the positive terminal and *exit* the negative terminal of an element. Below is an example for a resistor:



As long as this convention is followed consistently, it does not matter which direction you arbitrarily assigned each element current to; the voltage referencing will work out to determine the correct final sign. When we discuss *power* later in the module, you will see why we call this convention “passive.”

To achieve your goal of *finding* V_{out} , perform the rest of the NVA steps in the boxes below:

Step 5: Write KCL equations for all nodes with unknown voltages.

Step 6: Find expressions for all element currents in terms of element voltages and characteristics.

Step 7: Substitute all element voltages in your step 6 equations with node voltages.

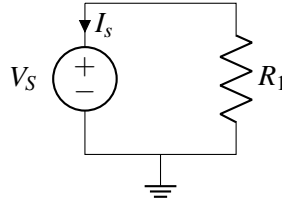
Step 8: Substitute expressions found in step 7 into the KCL equations from step 5.

Step 9: Solve for the node voltage values. At this point the analysis procedure is effectively complete - all that's left to do is solve the system of linear equations (by applying Gaussian Elimination, inverting $\mathbf{A}$, direct substitution, etc.) to find the values for the u 's. Then we can go back to our Step 7 equations and calculate the I 's. Note that in our circuit, $V_{out} = u_2 - 0 = u_2$.

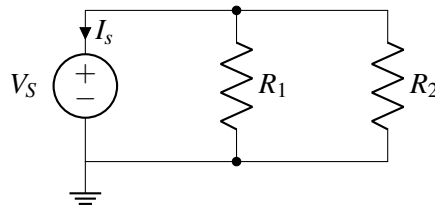
2. A Simple Circuit

Use KVL and/or KCL to solve the following circuits.

- (a) For this problem assume $V_S = 1V$ and $R_1 = 1k\Omega$. Find the current, I_s flowing through the voltage source.



- (b) For this problem assume $V_S = 1V$, $R_1 = 2k\Omega$, and $R_2 = 2k\Omega$. Find the current, I_s flowing through the voltage source.



EECS 16A Designing Information Devices and Systems I

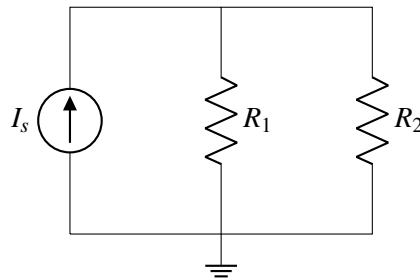
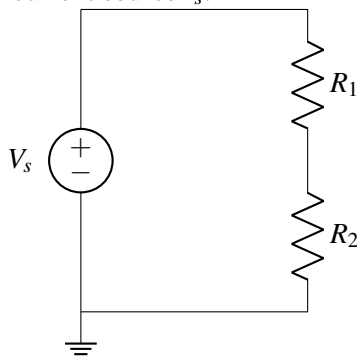
Discussion 7B

1. Current Divider

So far we've shown that for the circuit below to the left, the voltage across the resistor R_2 is expressed

$$V_{R_2} = \left(\frac{R_2}{R_1 + R_2} \right) V_s.$$

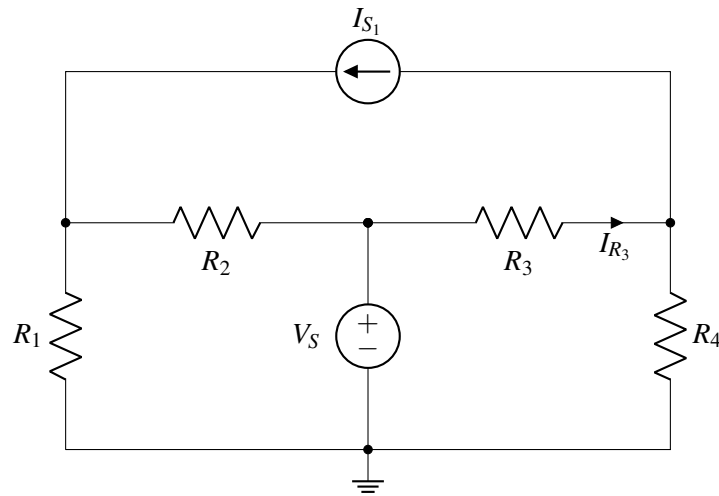
Let us derive a similar formula for the right circuit below, relating the current I_{R_2} through R_2 to the current through the current source I_s .



2. Circuit Analysis

Provided the circuit below...

- (a) ...use nodal analysis to solve for all node voltages.
- (b) ...find the current I_{R_3} flowing through resistor R_3 .



3. Bio-Molecule Detector

One application for electronics that has gained a lot of attention over the past several years is in so-called “bio-molecule” detection. The idea is to build a system that detects the presence of specific molecules and/or cells (e.g. specific viruses, proteins, etc.) in a biological sample; if this detection can be performed automatically and using relatively low-cost components, it can have a dramatic impact on a number of areas such as medical diagnosis, drug development, DNA sequencing, etc.

In this problem, we’ll look at how some of the techniques we learned about in the touchscreen module can be applied to realize a hypothetical bio-molecule detector. (Real bio-molecule detection systems involve quite a bit more complexity than what we’ll include here, but in many designs the same basic principles apply.)

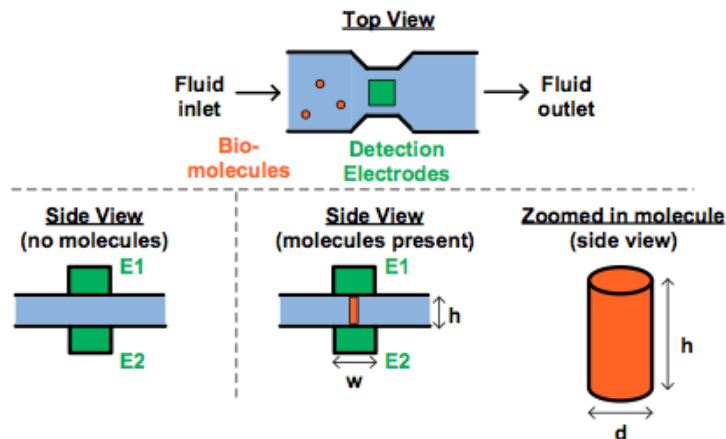


Figure 1: Bio-molecule detector.

As shown in Figure 1, the detector works by flowing a liquid that may or may not contain the biomolecules through a region in the device that has electrodes on the top and bottom of the liquid channel. The electrodes (E1/E2 in Figure 1) are chemically “functionalized” (using e.g. some appropriately designed antibodies), so that if the specific bio-molecule of interest is present in the fluid sample, one or more of the molecules will get physically trapped between the two electrodes (bottom right of Figure 1). After all of the fluid has been cleared out of the device (i.e., so that if there are no bio-molecules present, there is only air in between the two electrodes E1/E2), we can then figure out whether or not one or more bio-molecules were trapped by measuring the resistance between the two electrodes.

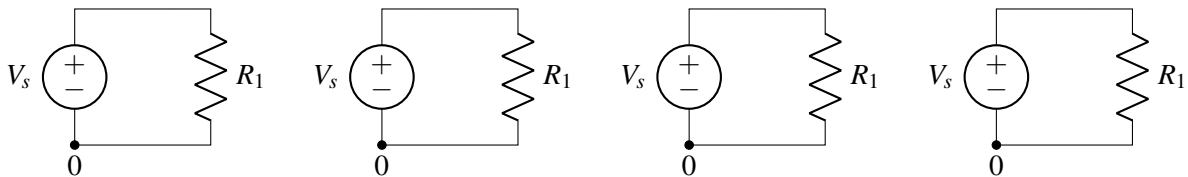
- Let’s first assume that we want to detect the presence of a bio-molecule by measuring resistance. If no bio-molecule is present, what should be the resistance between E1/E2? As shown in Figure 1, if each bio-molecule is a cylinder with diameter $d = 10\text{nm}$, height $h = 100\text{nm}$, and has a resistivity $\rho = 100\mu\Omega\text{m}$, what would be the resistance between E1 and E2 if only a single bio-molecule has been trapped? Note that you can assume that the trapped molecule is exactly vertically oriented when it is trapped – i.e., the top and bottom faces of the molecule are both aligned with surfaces of the electrodes.
- Using the same numbers for d , h , and ρ as part (a), as a function of the number of trapped bio-molecules $N_{\text{molecules}}$, what is the resistance between E1 and E2? (Note that you can assume that $N_{\text{molecules}}$ is small enough that all of the molecules fit within the electrode area and that all of the molecules are still trapped in an exactly vertical orientation.)
- Given your answers to parts (a) and (b), design a circuit that will output a voltage greater than 2.5 V if more than 5 molecules are trapped.

EECS 16A Designing Information Devices and Systems I

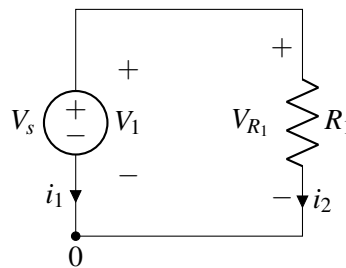
Discussion 8A

1. Passive Sign Convention and Power

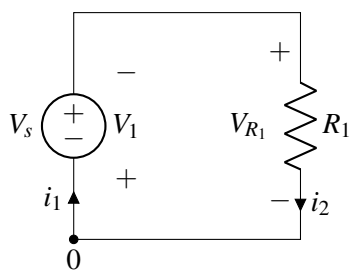
- (a) Below are four copies of the same single-resistor circuit. On each copy, provide a distinct choice of labels for each circuit's voltage polarities and current directions (there should be 4 possible choices in total!) while keeping with passive sign convention.



- (b) Suppose we consider one of the possible labelings you have found above (shown below). Calculate the power dissipated by every element in the circuit. Let $V_s = 5\text{ V}$ and let $R_1 = 5\ \Omega$. Recall that the power dissipated is the rate of electric energy converted into other forms and is given by the equation $P = IV$. When the power dissipated by an element is a negative value, it signifies that element is actually supplying electrical power to the circuit.



- (c) Suppose we choose a second labeling of the circuit as shown below. Calculate the power dissipated by every element in the circuit. Let $V_s = 5\text{ V}$ and let $R_1 = 5\ \Omega$.



- (d) Did the values of the element voltages and element currents change with the different labeling? Did the power for each circuit element change? Did the node voltages change? If a quantity didn't change with a difference in labeling, discuss what would have to change for quantity to change.

2. Volt and ammeter

Consider the following circuit below. We have also included relevant NVA equations below it.

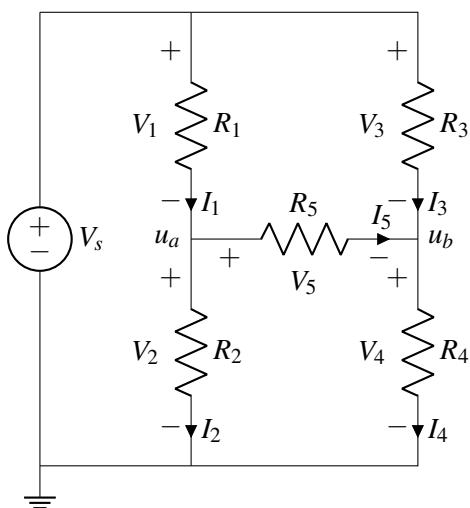


Figure 1: Circuit consisting of a voltage source V_s and five resistors R_1 to R_5 .

$$I_1 = I_2 + I_5$$

$$I_5 + I_3 = I_4$$

KCL

$$I_1 = \frac{V_1}{R_1} = \frac{V_s - u_a}{R_1}$$

$$I_2 = \frac{V_2}{R_2} = \frac{u_a - 0}{R_2}$$

$$I_3 = \frac{V_3}{R_3} = \frac{V_s - u_b}{R_3}$$

$$I_4 = \frac{V_4}{R_4} = \frac{u_b - 0}{R_4}$$

$$I_5 = \frac{V_5}{R_5} = \frac{u_a - u_b}{R_5}$$

Ohm's law in terms of node voltages

$$\frac{V_s - u_a}{R_1} = \frac{u_a - 0}{R_2} + \frac{u_a - u_b}{R_5}$$

$$\frac{u_a - u_b}{R_5} + \frac{V_s - u_b}{R_3} = \frac{u_b - 0}{R_4}$$

Substitute Ohm's into KCL

- (a) The circuit diagram shown in Figure 1 has been redrawn in Figure 2 by adding a voltmeter (letter V in a circle and plus and minus signs indicating direction) to measure voltage $V_{ab} = u_a - u_b$. Assume that the voltmeter is ideal. Are the values of V_{ab} before adding the voltmeter and after adding the voltmeter different? If so, which of the given equations change when doing NVA?

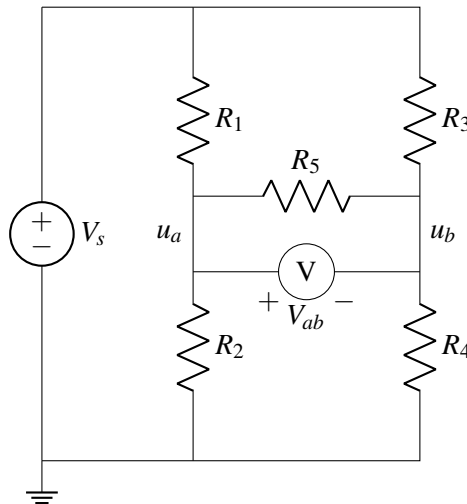


Figure 2: Circuit with voltmeter.

- (b) Suppose you accidentally connect an ideal ammeter in part (a) to nodes u_a and u_b instead of an ideal voltmeter. Does the value of V_{ab} with the ammeter connected differ from the value of V_{ab} without the ammeter connected? If so, what equations change when doing NVA?

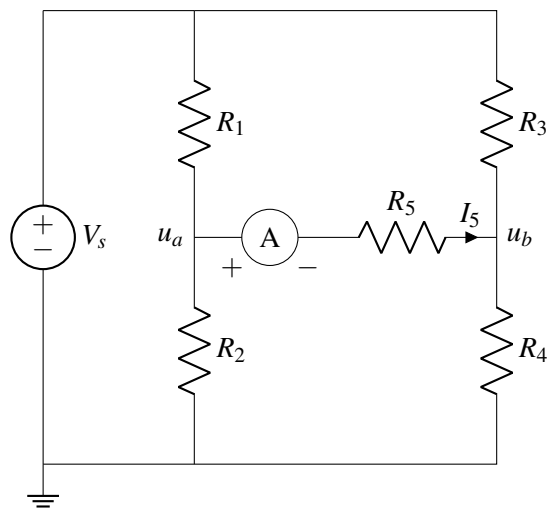


Figure 3: Circuit with ammeter.

- (c) The circuit diagram shown in Figure 1 has been redrawn in Figure 3 by adding an ideal ammeter (letter A in a circle and plus and minus signs indicating direction) in series with resistor R_5 . This will measure the current I_5 through R_5 . Are the values of I_5 before adding the ammeter and after adding the ammeter different? If so, what equations change when doing NVA?
- (d) Your friend accidentally connects a voltmeter in part (c) above, rather than an ammeter. Are the values of I_5 before adding the ammeter and after adding the ammeter different? If so, what equations change when doing NVA?

3. Cell Phone Battery

As great as smartphones are, one of their drawbacks is that their batteries don't last a long time. For example, a Google Pixel phone, under typical usage conditions (internet, a few cat videos, etc.) uses 0.3W. We will model the battery as an ideal voltage source (which maintains a constant voltage across its terminals regardless of current) except that we assume that the voltage drops abruptly to zero when the battery is discharged (in reality the voltage drops gradually, but let's keep things simple).

Battery capacity is specified in mAh, which indicates how many mA of current the battery can supply for one hour before it needs to be recharged. The Pixel's battery has a battery capacity of 2770mAh at 3.8V. For example, this battery could provide 1000mA (or 3.8W) for 2.77 hours before the voltage abruptly drops from 3.8V to zero.

- (a) How long will a Pixel's full battery last under typical usage conditions?

- (b) How many coulombs of charge does the battery contain? How many usable electrons worth of charge are contained in the battery when it is fully charged? (An electron has $1.602 \times 10^{-19} \text{ C}$ of charge.)
- (c) Suppose the cell phone battery is completely discharged and you want to recharge it completely. How much energy (in J) is this? Recall that a J is equivalent to a Ws.
- (d) Suppose PG&E charges \$0.12 per kWh. Every day, you completely discharge the battery (meaning more than typical usage) and you recharge it every night. How much will recharging cost you for the month of October (31 days)?
- (e) The battery has internal circuitry that prevents it from getting overcharged (and possibly exploding!). We will model the battery and its internal circuitry as a resistor R_{bat} . We now wish to charge the battery by plugging into a wall plug. The wall plug can be modeled as a 5 V voltage source and $200 \text{ m}\Omega$ resistor, as pictured in Figure 4. What is the power dissipated across R_{bat} for $R_{\text{bat}} = 1 \text{ m}\Omega$, 1Ω , and $10 \text{ k}\Omega$? (i.e. how much power is being supplied to the phone battery as it is charging?). How long will the battery take to charge for each of those values of R_{bat} ?

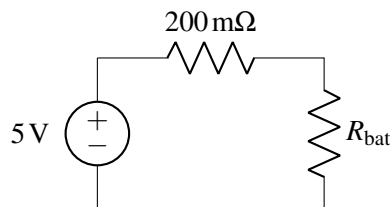


Figure 4: Model of wall plug, wire, and battery.

EECS 16A Designing Information Devices and Systems I

Discussion 8B

1. Resist the Touch

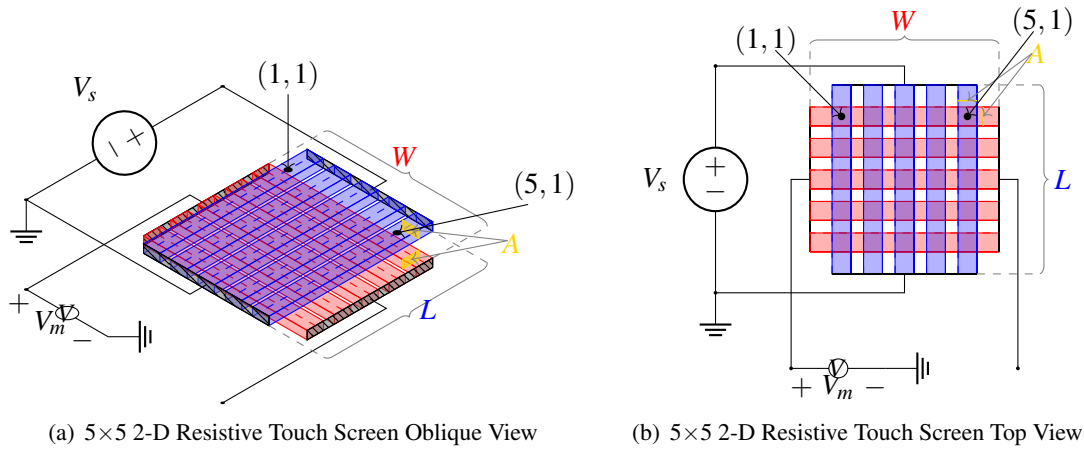


Figure 1: $N \times N$ Resistive Touch Screen, $N = 5$

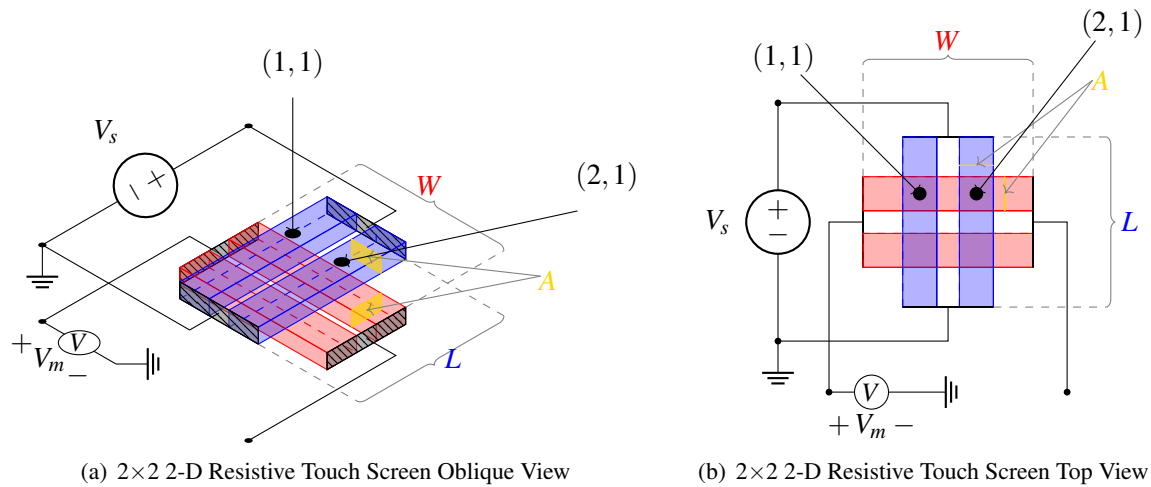
In this question we will be re-examining the 2-dimensional resistive touchscreen. This touchscreen, is slightly different to the one shown in lecture and more like the one we will be examining in lab.

The touchscreen has length L and width W and is composed of a rigid bottom-layer and a flexible top-layer. Instead of having two continuous resistive sheets on the top and bottom layers, this is a simpler implementation with N vertical strips of conductive material in the top layer and N horizontal strips of conductive material in the bottom layer. The strips of a single layer are all connected by an ideal conducting plate on each side. All strips have resistivity, ρ , and cross-sectional area, A .

Assume that all top layer resistive strips and bottom layer resistive strips are spaced apart equally, and that the upper left touch point in Figure 1(b) is position $(1, 1)$, and the upper right touch point is $(N, 1)$. The spacing (measured from the center of one strip to the center of the next) between the strips in the top layer is $\frac{W}{N+1}$, and the spacing between the strips in the bottom layer is $\frac{L}{N+1}$.

- (a) Find the resistance R_y for a single vertical blue strip and R_x for a single horizontal red strip, as a function of the screen dimensions W and L , the strip resistivity ρ , and the cross-sectional area A .

- (b) Consider a 2×2 example for the touchscreen circuit, shown in Figure 2.

Figure 2: 2×2 Resistive Touch Screen

Assume that we connect a voltage source V_s , between the top and bottom terminals of the blue strips, and a voltmeter V_m to one of the left or right terminals as depicted in the diagram.

If $V_s = 3\text{ V}$, $R_x = 2000\Omega$, and $R_y = 2000\Omega$, draw the equivalent circuit for when the point $(2,2)$ is pressed and solve for the measured voltage, V_m , with respect to ground.

Remember that all top layer resistive strips and bottom layer resistive strips are spaced apart equally, and that the upper left touch point is position $(1,1)$. The spacing between the strips in the top layer is $\frac{W}{N+1}$, and the spacing between the strips in the bottom layer is $\frac{L}{N+1}$.

- (c) Suppose a touch occurs at coordinates (i, j) for an arbitrary $N \times N$ touchscreen, and the voltage source and meter are connected as in the figures. A 5×5 example is shown in Figure 1(b). Find an expression for V_m as a function of V_s , N , i , and j . Again, the upper left corner is the coordinate $(1, 1)$ and the upper right coordinate is $(N, 1)$

- (d) Optional / Fun: Experiment with the TinkerCad models below to validate the theoretical results you just derived.

TinkerCad model of 2×2 equivalent circuit: <https://www.tinkercad.com/things/0wIXz3MkD7B>

TinkerCad model of 3×2 equivalent circuit: <https://www.tinkercad.com/things/k5oolj2tUEN>

2. Cell Phone Battery

As great as smartphones are, one of their drawbacks is that their batteries don't last a long time. For example, a Google Pixel phone, under typical usage conditions (internet, a few cat videos, etc.) uses 0.3W. We will model the battery as an ideal voltage source (which maintains a constant voltage across its terminals regardless of current) except that we assume that the voltage drops abruptly to zero when the battery is discharged (in reality the voltage drops gradually, but let's keep things simple).

Battery capacity is specified in mAh, which indicates how many mA of current the battery can supply for one hour before it needs to be recharged. The Pixel's battery has a battery capacity of 2770mAh at 3.8V. For example, this battery could provide 1000mA (or 3.8W) for 2.77 hours before the voltage abruptly drops from 3.8V to zero.

- (a) How long will a Pixel's full battery last under typical usage conditions?

- (b) How many coulombs of charge does the battery contain? How many usable electrons worth of charge are contained in the battery when it is fully charged? (An electron has 1.602×10^{-19} C of charge.)

- (c) Suppose the cell phone battery is completely discharged and you want to recharge it completely. How much energy (in J) is this? Recall that a J is equivalent to a W s.
- (d) Suppose PG&E charges \$0.12 per kWh. Every day, you completely discharge the battery (meaning more than typical usage) and you recharge it every night. How much will recharging cost you for the month of October (31 days)?
- (e) The battery has internal circuitry that prevents it from getting overcharged (and possibly exploding!). We will model the battery and its internal circuitry as a resistor R_{bat} . We now wish to charge the battery by plugging into a wall plug. The wall plug can be modeled as a 5 V voltage source and $200\text{ m}\Omega$ resistor, as pictured in Figure 3. What is the power dissipated across R_{bat} for $R_{\text{bat}} = 1\text{ m}\Omega$, $1\text{ }\Omega$, and $10\text{ k}\Omega$? (i.e. how much power is being supplied to the phone battery as it is charging?). How long will the battery take to charge for each of those values of R_{bat} ?

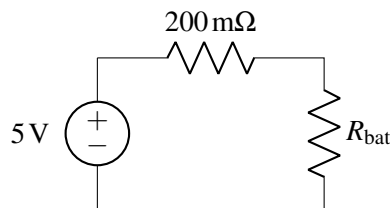


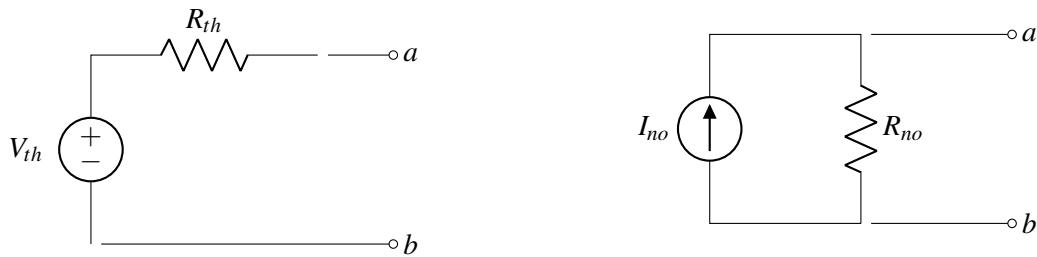
Figure 3: Model of wall plug, wire, and battery.

EECS 16A Designing Information Devices and Systems I

Discussion 9A

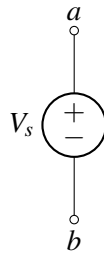
1. Thevenin and Norton Equivalence

The general Thévenin and Norton equivalents are shown below:

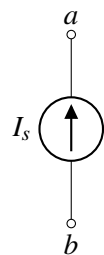


Find the Thévenin and Norton equivalents across terminals a and b for the circuits given below.

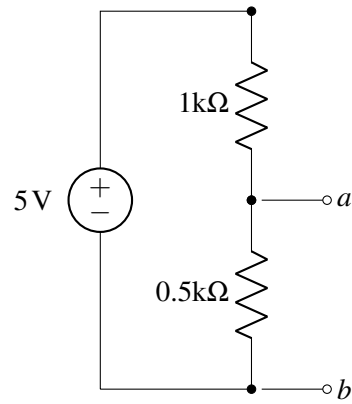
(a)



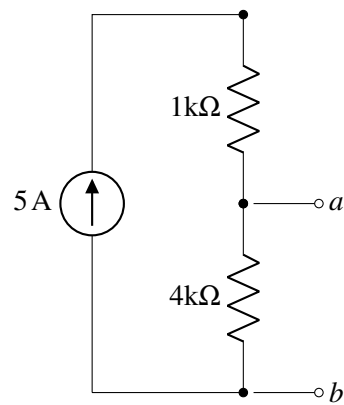
(b)



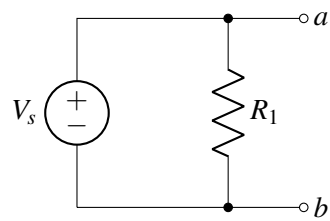
(c)



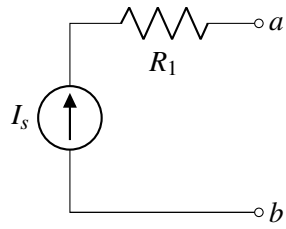
(d)



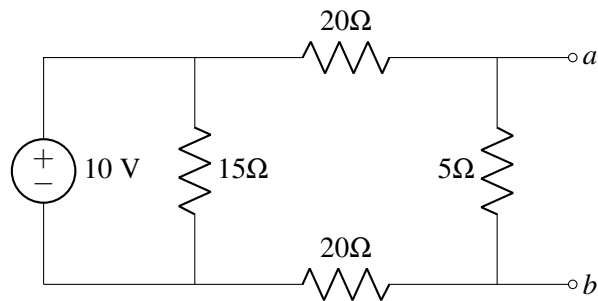
(e)



(f)

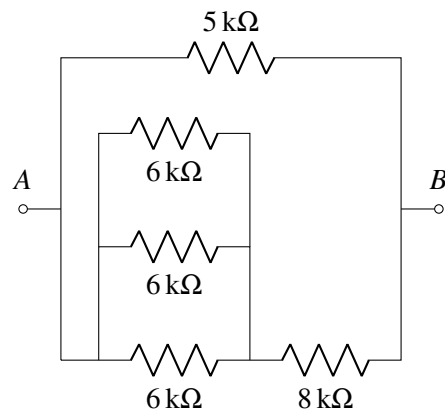


(g)



2. Series and Parallel Combinations

For the resistor network shown below, find an equivalent resistance between the terminals A and B using the resistor combination rules for series and parallel resistors.



EECS 16A Designing Information Devices and Systems I

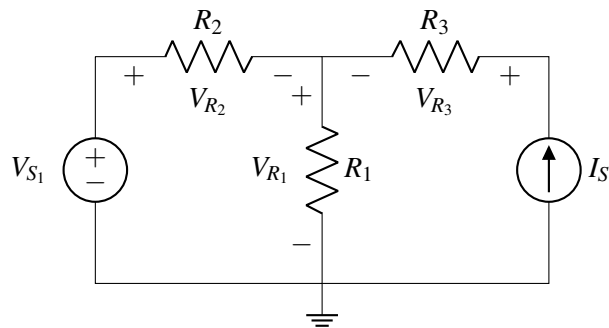
Discussion 9B

1. Superposition

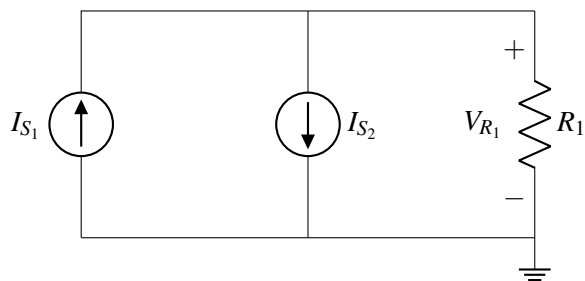
For the following circuits:

- Use the superposition theorem to solve for the voltages across the resistors.
- For parts (b) and (c) only, find the power dissipated/generated by all components. Is power conserved?

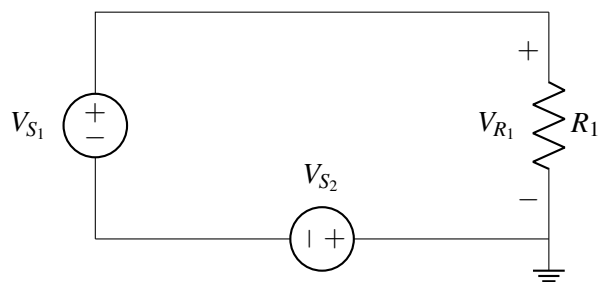
(a)



(b)

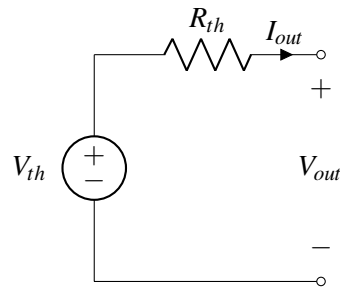
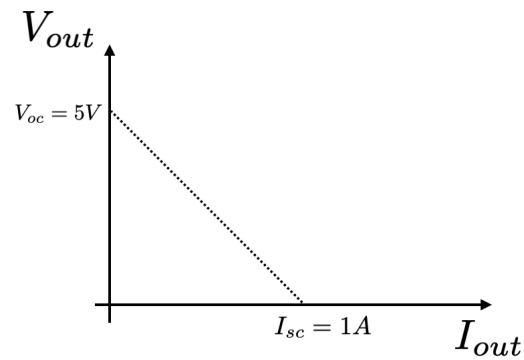


(c)

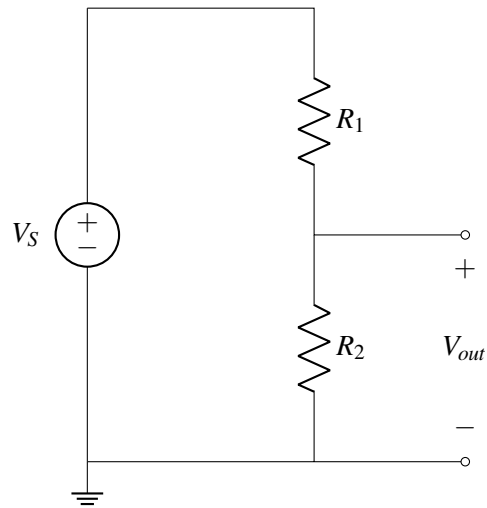


2. Thevenin equivalence

- (a) You are given the following $I_{out} - V_{out}$ characteristic of the Thevenin model of a circuit. Find the Thevenin voltage and the Thevenin resistance.



- (b) You are given a voltage divider as shown below. Find R_1 and R_2 such that the Thevenin equivalent model is the same as that of (a). You are given that $V_S = 10V$.



EECS 16A Designing Information Devices and Systems I

Discussion 10A

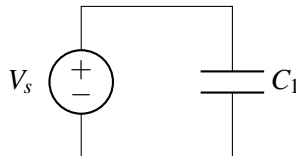
1. Voltages Across Capacitors

For the circuits given below, determine the voltage across each capacitor and calculate the charge and energy stored on each capacitor (assume all capacitors start *uncharged*, and then we've let the system reach steady state). We are also given $C_1 = 1\ \mu\text{F}$, $C_2 = 3\ \mu\text{F}$, and $V_s = 1\ \text{V}$.

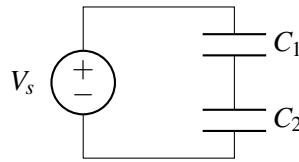
Recall charge has units of Coulombs (C), and capacitance is measured in Farads (F) = $\frac{\text{Coulomb}}{\text{Volt}}$.

It may also help to note metric prefix examples: $3\ \mu\text{F} = 3 \times 10^{-6}\text{F}$.

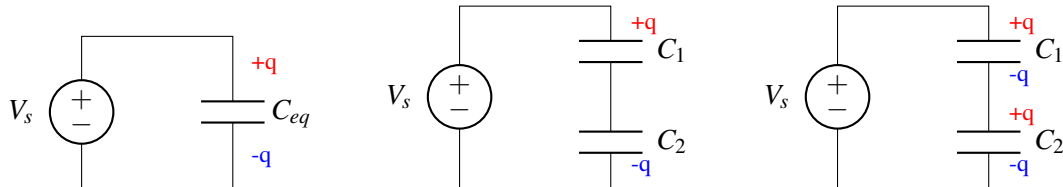
(a)



(b)



Helpful diagrams for considering the charges capacitors linked in series:
(without any initial charges)



Left: Our series capacitors may be modeled as one equivalent capacitor C_{eq} , which after some time is charged up by V_s to have $+q$ on the top plate and $-q$ on the bottom plate.

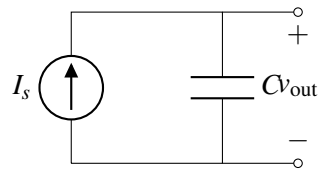
Middle: We return to the 2-capacitor picture, but carry this insight of equivalent charge with us. Now the charge $+q$ is on the top plate of capacitor C_1 , and $-q$ is on the bottom plate of capacitor C_2 .

Right: Since capacitor plates have opposite & equal charges, we attain this final right diagram.

As another conceptual check, we notice that the node between C_1 and C_2 is isolated from any other connections and should always remain *charge neutral*. From the diagram right we see this is maintained since $(+q) + (-q) = 0$.

2. Current Sources And Capacitors

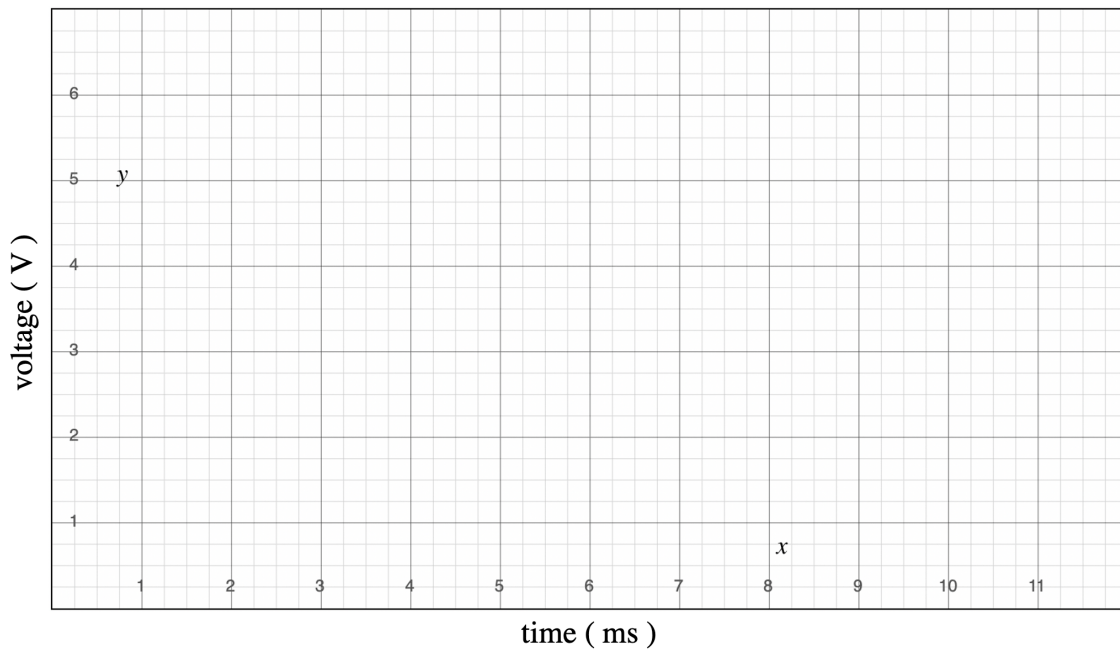
Given the circuit below, find an expression for $v_{\text{out}}(t)$ in terms of I_s , C , V_0 , and t , where V_0 is the initial voltage across the capacitor at $t = 0$.



Then plot the function $v_{\text{out}}(t)$ over time on the graph below for the following conditions detailed below. Use the values $I_s = 1\text{mA}$ and $C = 2\mu\text{F}$.

- Capacitor is initially uncharged $V_0 = 0$ at $t = 0$.
- Capacitor has been charged with $V_0 = +1.5\text{V}$ at $t = 0$.
- Practice:** Swap this capacitor for one with half the capacitance $C = 1\mu\text{F}$, which is initially uncharged $V_0 = 0$ at $t = 0$.

HINT: Recall the calculus identity $\int_a^b f'(x)dx = f(b) - f(a)$, where $f'(x) = \frac{df}{dx}$.

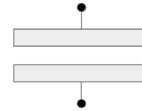


3. Additional reading: capacitance and equivalence

This serves as an aid for students to better master parallel plate capacitors and their equivalence formulae.

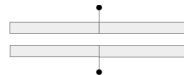
In this course we discuss *parallel-plate* capacitors, which are two closely-spaced metal plates. Additionally we provide the equation for its capacitance:

$$C = \epsilon \frac{A}{d}$$



Here the plates have an area of $A = L \times W$ (where L is length going into the page and W is the width) and are spaced a distance d apart.

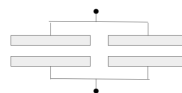
- (a) Let's consider a capacitor with the same L and d dimensions as the capacitor described above, yet with twice the width $2W$. This is illustrated below



Since the capacitor plates have been doubled in area, the new capacitance is

$$C = \epsilon \frac{(2W) \times L}{d} = 2 \left(\epsilon \frac{A}{d} \right).$$

- (b) Let's next consider two capacitors each with identical dimensions as in the first example (W, L, d) arranged in parallel as shown below. With the exercise above in mind, what do you think the capacitance of this circuit is?



Intuitively, this should agree with our expression found above, since both systems appear as one capacitor with twice the plate area. Thus, the answer remains $C_{eq} = 2 \left(\epsilon \frac{A}{d} \right)$.

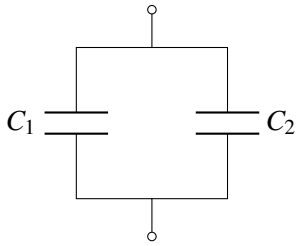
EECS 16A Designing Information Devices and Systems I

Discussion 10B

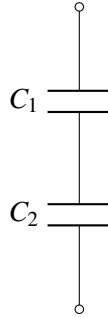
1. Practice: Series And Parallel Capacitors

Derive C_{eq} for the following circuits.

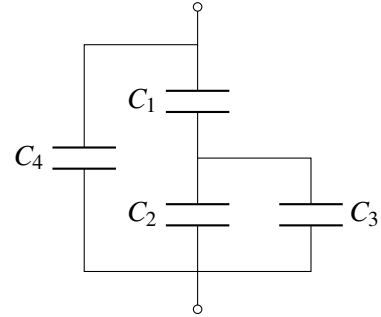
(a)



(b)

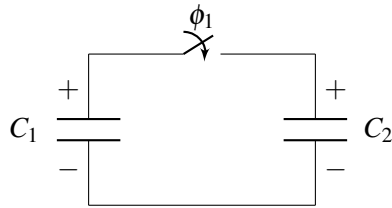


(c)

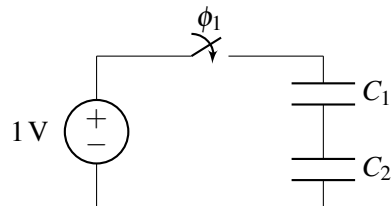


2. Capacitors and Charge Conservation

- (a) Consider the circuit below with $C_1 = C_2 = 1\ \mu\text{F}$ and an open switch. Suppose that C_1 is initially charged to $+1\ \text{V}$ and that C_2 is charged to $+2\ \text{V}$. How much charge is on C_1 and C_2 ?



- (b) Now the switch is closed (i.e. the capacitors are connected together.) What are the voltages across and the charges on C_1 and C_2 ?
- (c) Consider the following circuit with $C_1 = 1\ \mu\text{F}$ and $C_2 = 3\ \mu\text{F}$. Suppose that both capacitors are initially uncharged ($0\ \text{V}$).



What are the voltages across each capacitor after the switch is closed? What are the charges on each capacitor?

3. Bio-Molecule Detector [PRACTICE]

We've already seen how to build a bio-molecule detector where bio-molecules change the resistance between two electrodes. In this problem, we will explore a different scheme, where bio-molecules change the capacitance between two electrodes.

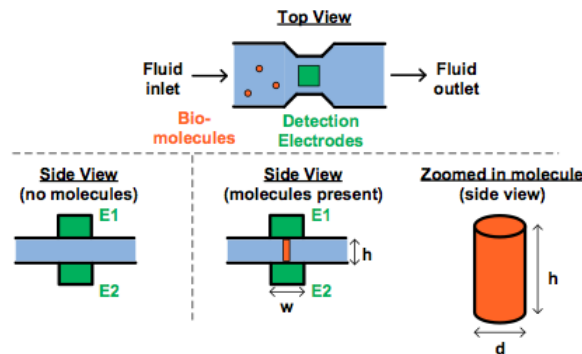


Figure 1: Bio-molecule detector.

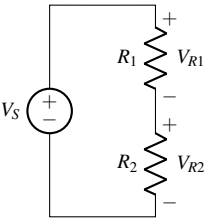
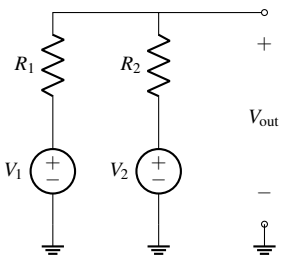
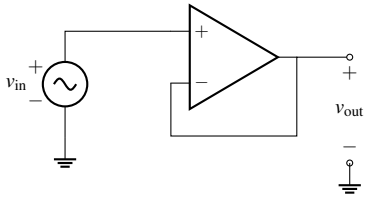
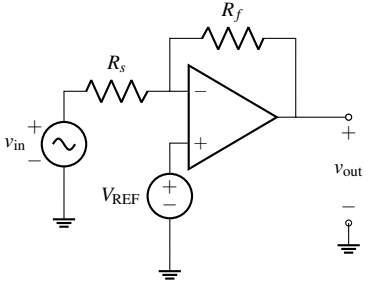
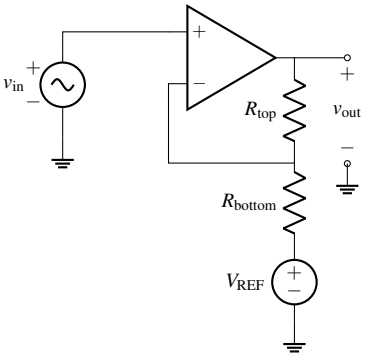
As shown in Figure 1, the detector works by flowing a liquid that may or may not contain the biomolecules through a region in the device that has electrodes on the top and bottom of the liquid channel. The electrodes (E1/E2 in Figure 1) are chemically “functionalized” (using e.g. some appropriately designed antibodies), so that if the specific bio-molecule of interest is present in the fluid sample, one or more of the molecules will get physically trapped between the two electrodes (bottom right of Figure 1). After all of the fluid has been cleared out of the device (i.e., so that if there are bio-molecules present, there is only air in between the two electrodes E1/E2), we can then figure out whether or not one or more bio-molecules were trapped by measuring the capacitance between the two electrodes.

- If no bio-molecules are present, what should the capacitance between E1/E2 be? Assume that the electrodes are $10\mu\text{m}$ on each side and that the electrodes are 100nm apart. Leave your answer in terms of ϵ_0 .
- As shown in Figure 1, if each bio-molecule is a cylinder with diameter $d = 2\mu\text{m}$, height $h = 100\text{nm}$, and permittivity $\epsilon_b = 4\epsilon_0$, what would the capacitance between E1 and E2 be if only a single bio-molecule is trapped? Note that you can assume that the trapped molecule is exactly vertically oriented when it is trapped – i.e., the top and bottom faces of the molecule are both aligned with surfaces of the electrodes.
- Using the same numbers for d , h , and ϵ_b as in part (b), as a function of the number of trapped bio-molecules $N_{\text{molecules}}$, what is the capacitance between E1 and E2? (Note that you can assume that $N_{\text{molecules}}$ is small enough that all of the molecules fit within the electrode area and that all of the molecules are still trapped in exactly vertical orientation.)

EECS 16A Designing Information Devices and Systems I

Discussion 11A

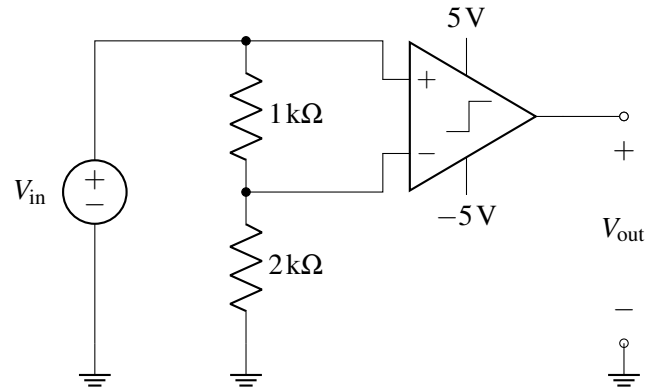
For Reference: Example Circuits

Voltage Divider  $V_{R2} = V_S \left(\frac{R_2}{R_1 + R_2} \right)$	Voltage Summer  $V_{out} = V_1 \left(\frac{R_2}{R_1 + R_2} \right) + V_2 \left(\frac{R_1}{R_1 + R_2} \right)$	Unity Gain Buffer  $\frac{v_{out}}{v_{in}} = 1$
Inverting Amplifier  $v_{out} = v_{in} \left(-\frac{R_f}{R_s} \right) + V_{REF} \left(\frac{R_f}{R_s} + 1 \right)$	Non-inverting Amplifier  $v_{out} = v_{in} \left(1 + \frac{R_{top}}{R_{bottom}} \right) - V_{REF} \left(\frac{R_{top}}{R_{bottom}} \right)$	

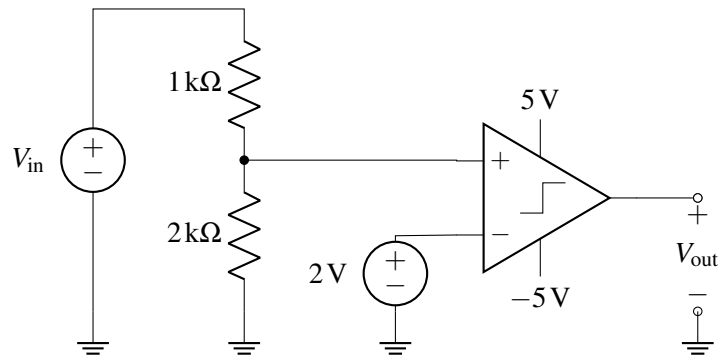
1. Comparators

For each of the circuits shown below, plot V_{out} for V_{in} ranging from -10 V to 10 V for part (a) and from 0 V to 10 V for part (b). Assume $V_{SS} = -V_{DD}$.

(a)

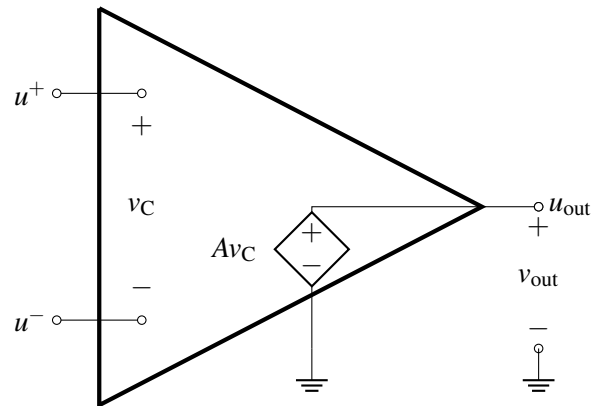


(b)



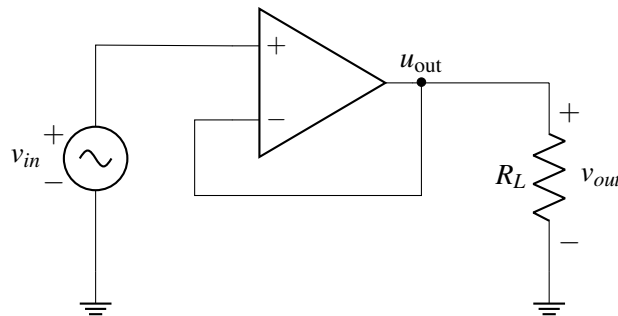
2. Op-Amp Rules and Negative Feedback Rules

Here is an equivalent circuit of an op-amp (where we are assuming that $V_{SS} = -V_{DD}$) for reference:



- What are the currents flowing into the positive and negative terminals of the op-amp (i.e., what are I^+ and I^-)? Based on this answer, what are some of the advantages of using an op-amp in your circuit designs?
- Suppose we add a resistor of value R_L between u_{out} and ground. What is the value of v_{out} ? Does your answer depend on R_L ? In other words, how does R_L affect Av_C ? What are the implications of this with respect to using op-amps in circuit design?

For the rest of the problem, consider the following op-amp circuit in negative feedback:

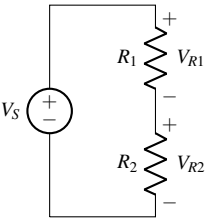
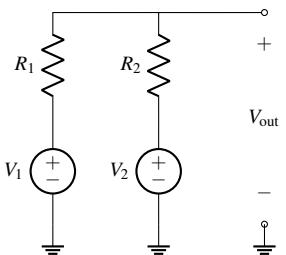
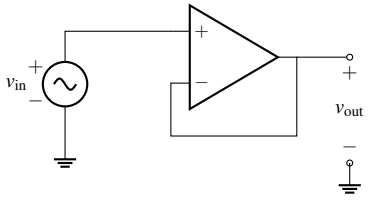
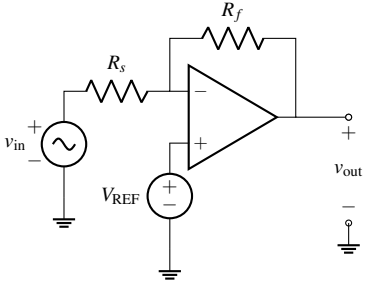
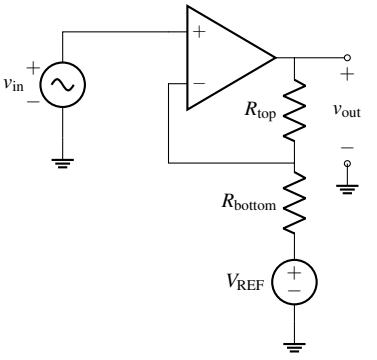


- Assuming that this is an ideal op-amp, what is v_{out} ?
- Draw the equivalent circuit for this op-amp and calculate v_{out} in terms of A , v_{in} , and R_L for the circuit in negative feedback. Does v_{out} depend on R_L ? What is v_{out} in the limit as $A \rightarrow \infty$?

EECS 16A Designing Information Devices and Systems I

Discussion 11B

For Reference: Example Circuits

Voltage Divider  $V_{R2} = V_S \left(\frac{R_2}{R_1 + R_2} \right)$	Voltage Summer  $V_{out} = V_1 \left(\frac{R_2}{R_1 + R_2} \right) + V_2 \left(\frac{R_1}{R_1 + R_2} \right)$	Unity Gain Buffer  $\frac{v_{out}}{v_{in}} = 1$
Inverting Amplifier  $v_{out} = v_{in} \left(-\frac{R_f}{R_s} \right) + V_{REF} \left(\frac{R_f}{R_s} + 1 \right)$	Non-inverting Amplifier  $v_{out} = v_{in} \left(1 + \frac{R_{top}}{R_{bottom}} \right) - V_{REF} \left(\frac{R_{top}}{R_{bottom}} \right)$	

1. Noisy Parties (from Fa18 MT2)

To celebrate the (peaceful and very reasonable) end of the pizza-on-pineapple war, course staff decided to throw a party for the entire class! (Disclaimer: Not really) You and your friends want to take a picture to commemorate the night. In the imaging lab, you learned about how to make a camera with a single photodiode and a projector. Most modern cameras, however, have many photodiodes and do not need a sweeping light source like what you used in lab and need to work in non-ideal, non-laboratory conditions.

(a) We can model a photodiode as a capacitor in parallel with a current source:

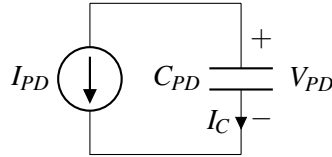


Figure 1: Photodiode circuit model

The current I_{PD} is related to the intensity of the light shining on the photodiode. Start with the following equations:

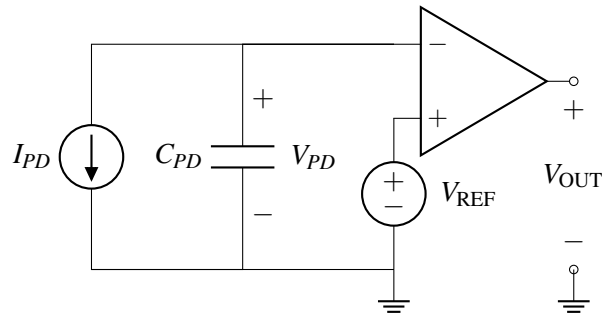
- $V_{PD}(t = 0) = V_{DD}$
- $I_C = C_{PD} \frac{dV_{PD}}{dt}$

Begin with the above equations and show that $V_{PD}(t) = V_{DD} - \frac{I_{PD}}{C_{PD}}t$.

Hint: Follow passive sign convention.

- (b) When feeding information from circuits into computers, we need to convert from our continuous-valued analog signals into discrete-valued digital signals the computer can understand using an analog-to-digital converter (ADC).

One type of ADC uses the time it takes for an event to occur in its conversion (converting time to the digital domain involves counting at a steady rate until the event occurs). We care about the time it takes for the comparator output to change.



Parameter	Unit	Value
I_{PD}	A	$1 \cdot 10^{-3}$
C_{PD}	F	$1 \cdot 10^{-9}$
V_{REF}	V	0.5
V_{DD}	V	1.0

Figure 2: Integrating ADC

Table 1: Numerical values for parts (b) and (c)

Draw $V_{PD}(t)$ in the graph below, where $V_{PD}(t = 0) = V_{DD}$. Clearly label the following with numerical values:

- Slope of the $V_{PD}(t)$ vs. t curve, i.e. $\frac{dV_{PD}}{dt}$.
- t_{flip} where $V_{PD}(t_{\text{flip}}) = V_{REF}$ and the comparator output goes high for the first time
- t_{zero} where $V_{PD}(t_{\text{zero}}) = 0V$

Hint: $1\mu\text{s} = 1 \cdot 10^{-6}\text{s}$.

- (c) Everyone in your picture is holding still, and in previous parts we assumed a constant current across the capacitor. However, someone just turned on strobe lights! You only want a picture of people, so we can treat the flashing lights as interference. Now our full circuit looks like Figure 3

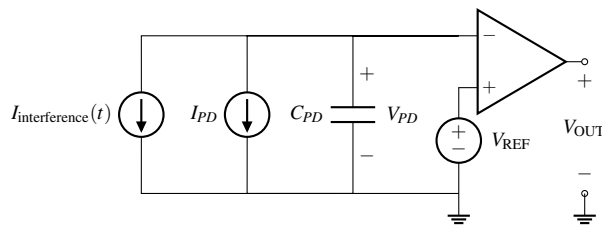


Figure 3: Model with interference

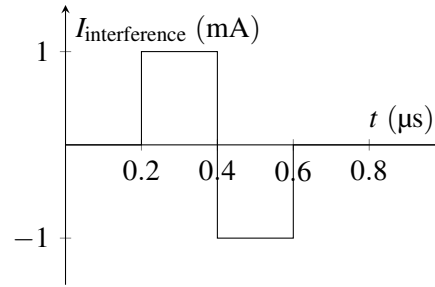


Figure 4: Interference current for part (c)

For this subpart, use the interference current defined in Figure 4 and the values in Table 1 (repeated for convenience).

Parameter	Unit	Value
I_{PD}	A	$1 \cdot 10^{-3}$
C_{PD}	F	$1 \cdot 10^{-9}$
V_{REF}	V	0.5
V_{DD}	V	1.0

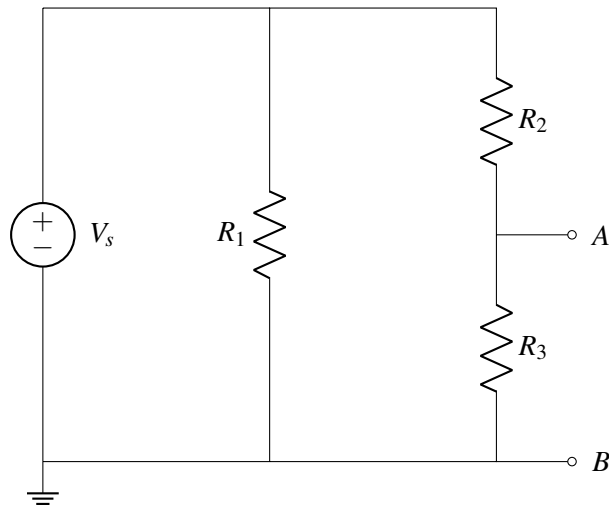
Table 1 repeated for convenience

Draw $V_{PD}(t)$ in the graph below given $V_{PD}(0) = V_{DD}$. Clearly label $\frac{dV_{PD}}{dt}$ for every line segment in your graph with numerical values. Also find t_{flip} where $V_{PD}(t_{\text{flip}}) = V_{REF}$.

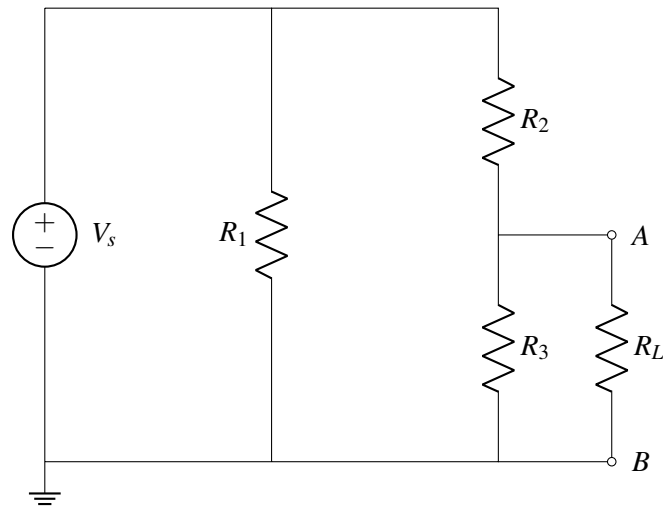
Hint: $V_{PD}(t)$ is continuous.

2. Thévenin/Norton Equivalence

- (a) Find the Thévenin resistance R_{th} of the circuit shown below, with respect to its terminals A and B .



- (b) Now, a load resistor, $R_L = R$, is connected across terminals A and B , as shown in the circuit below. Find the power dissipated in the load resistor in terms of the given variables.

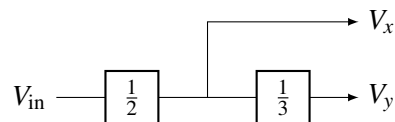


EECS 16A Designing Information Devices and Systems I

Discussion 12A

1. Modular Circuit Buffer

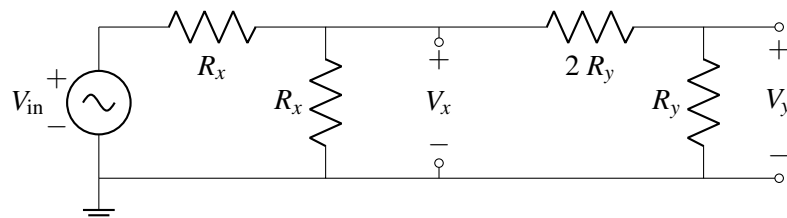
Let's try designing circuits that perform a set of mathematical operations using op-amps. While voltage dividers on their own cannot be combined without altering their behavior, op-amps can preserve their behavior when combined and thus are a perfect tool for modular circuit design. We would like to implement the block diagram shown below:



In other words, create a circuit with two outputs V_x and V_y , where $V_x = \frac{1}{2}V_{in}$ and $V_y = \frac{1}{3}V_x = \frac{1}{6}V_{in}$.

- (a) Draw two voltage dividers, one for each operation (the $1/2$ and $1/3$ scalings). What relationships hold for the resistor values for the $1/2$ divider, and for the resistor values for the $1/3$ divider?

- (b) If you combine the voltage dividers, made in part (a), as shown by the block diagram (output of the $1/2$ voltage divider becomes the source for the $1/3$ voltage divider circuit), do they behave as we hope (meaning $V_{in} = 2V_x = 6V_y$)?



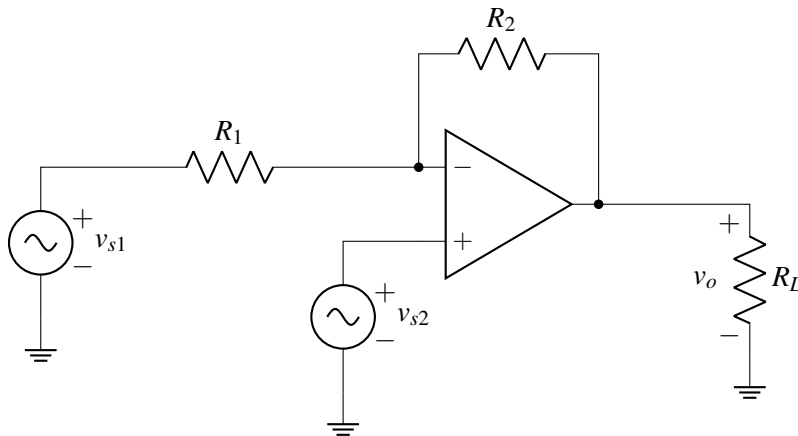
(c) Perhaps we could use an op-amp (in negative-feedback) to achieve our desired behavior.

Modify the implementation you tried in part (b) using a negative feedback op-amp in order to achieve the desired V_x, V_y relations $V_x = \frac{V_{in}}{2}$ and $V_y = \frac{V_x}{3} = \frac{V_{in}}{6}$.

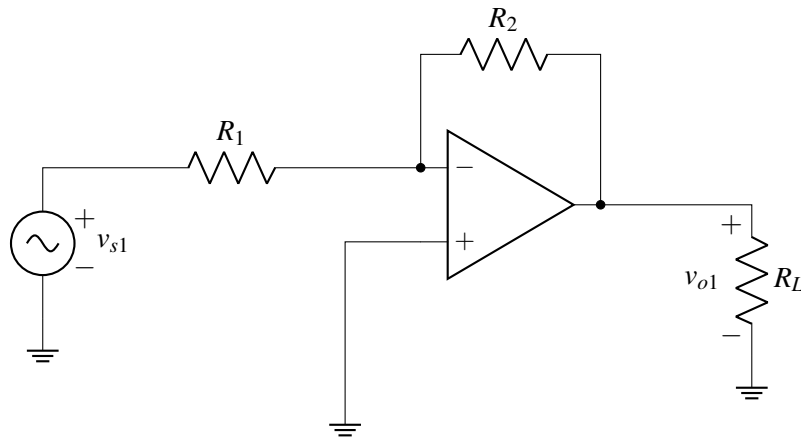
HINT: Place the op-amp in between the dividers such that the V_x node is an input into the op-amp, while the source of the 2nd divider is the output of the op-amp!

2. Amplifier with Multiple Inputs

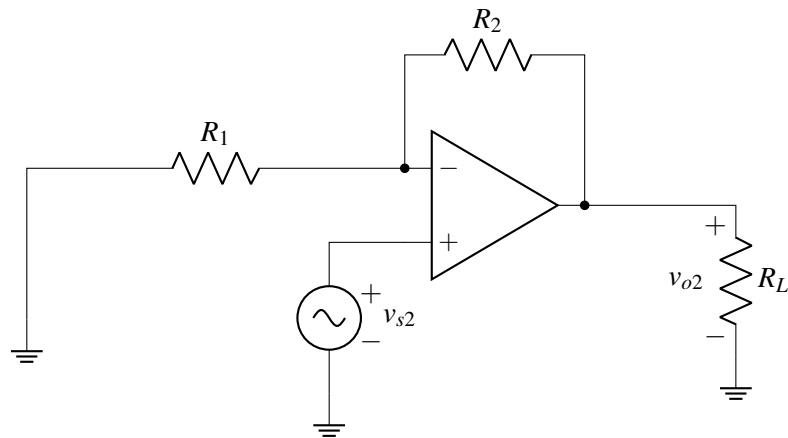
In this problem we will use superposition and the Golden Rules to find the output of the following op amp circuit with multiple inputs:



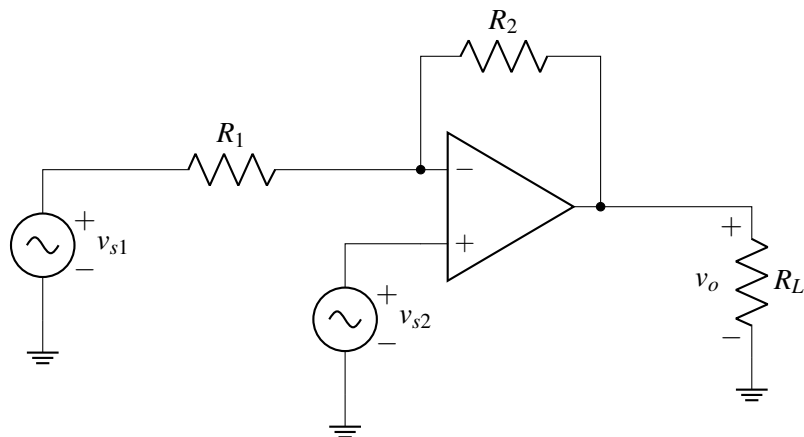
(a) First, let's turn off v_{s2} . Use the **Golden Rules** to find v_{o1} for the circuit below.



(b) Now let's turn off v_{s1} . Use the **Golden Rules** to find v_{o2} for the circuit below.



(c) Use **superposition** to find the output voltage v_o for the circuit shown below.



3. Modular Op-Amp Circuits

Let's design blocks that implement the following operations:

- (a) Scale the input voltage so that: $V_{\text{out}} = +5 V_{\text{in}}$.
- (b) Scale and invert the input voltage so that: $V_{\text{out}} = -2 V_{\text{in}}$.

EECS 16A Designing Information Devices and Systems I

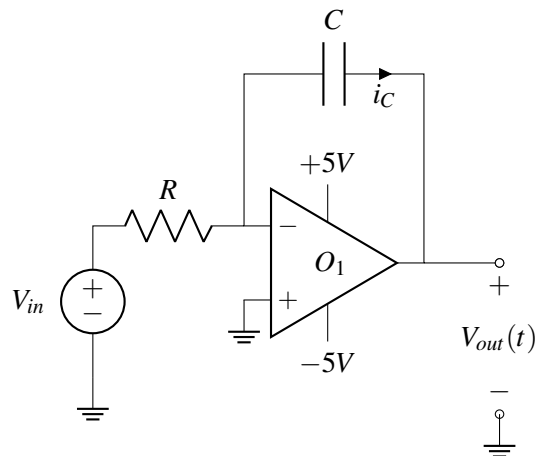
Discussion 12B

1. Integration using Op-amps

As we have seen already, it is useful in several applications to create triangular wave voltages (also known as voltage ramps). For example, in the capacitive touchscreen, we charged a capacitor using a constant current source that flipped from positive current to negative current periodically, which created triangular waveforms for the voltage on the capacitor. Then we compared the capacitor's voltage to a reference voltage to detect touch. In this problem, you will be analyzing a circuit that produces voltage ramps using a voltage source and an op-amp in negative feedback.

- (a) One of the circuit blocks you can use to generate the triangular waveform is the integrator. An integrator outputs the integral of the input signal. For the circuit given below express $V_{out}(t)$ in terms of R , C , V_{in} , and t , assuming V_{in} is not varying with time. What is the slope of $V_{out}(t)$? You may also assume that capacitor C has 0 V across it at time $t = 0$.

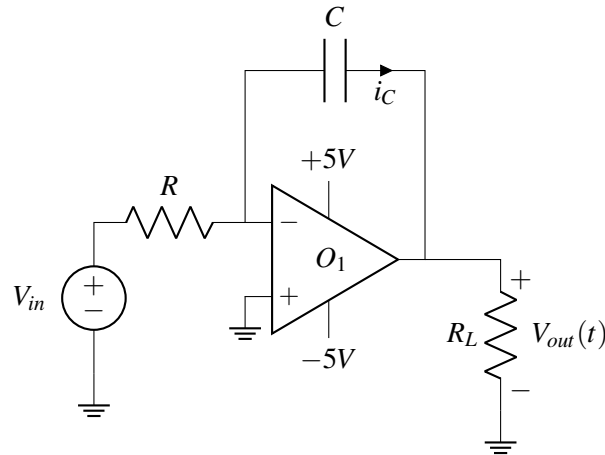
Hint: You will have to apply KCL, and use the fact that the current flowing through a capacitor is given by $I = C \frac{dV}{dt}$.



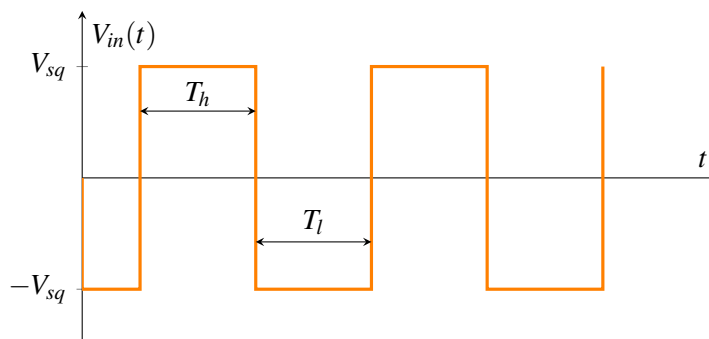
- (b) What is the value of the current i_C flowing through capacitor C , in terms of V_{in} , R and C ? How does the capacitor current change if we double C ? How does the slope of $V_{out}(t)$ change if we double capacitor C ?

Note: the current direction is specified in the figure above.

- (c) If we connect a load resistance at the output of the circuit, as shown in the figure below, does the output voltage $V_{out}(t)$ change, from what you calculated in part (a)? Why or why not?

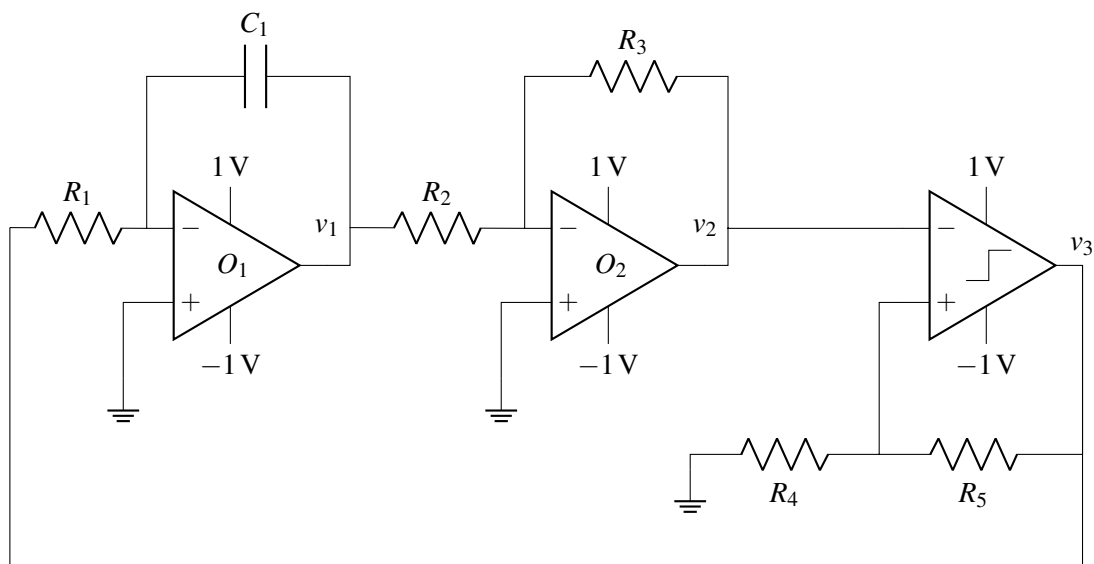


- (d) If $V_{in}(t)$ varies with time as shown in the following diagram, plot $V_{out}(t)$ for $t = 0$ to $t = 1.5T$, where $T = 2T_h = 2T_l$. In your plot indicate an algebraic expression for the slope (as a function of R , C and V_{sq}) and add tick marks on the x and y axis indicating the time and voltage values where the ramp slope changes. You may assume again that capacitor C has 0V across it at time $t = 0$.



2. Timer Circuit

In this problem, we will walk through another useful, real-world circuit, the timer circuit. The circuit is shown below. All resistors have a resistance of $1\text{ k}\Omega$ and $C_1 = 1\text{ }\mu\text{F}$. Assume that all the op-amps are in negative feedback.



- (a) Find the current through the capacitor C_1 in terms of the node voltage v_3 and the resistor R_1 .
- (b) Suppose that at time $t = 0$, C_1 is uncharged. Find the voltage v_1 in terms of t and v_3 . Recall that all resistors have a resistance of $1\text{ k}\Omega$ and $C_1 = 1\text{ }\mu\text{F}$. What is the maximum $|v_1|$ could be?
- (c) How is v_2 related to v_1 ? What is the voltage v_2 ?

Now, let's independently analyze the circuit in the two possible outputs of the comparator, when $v_3 = 1\text{ V}$ and when $v_3 = -1\text{ V}$.

- (d) Assume that the output of the comparator v_3 has railed to the top rail. With this value of v_3 , what is v_2 as a function of time? What is the voltage v_c^+ at the positive input of the comparator? At what time will the two inputs of the comparator be equal?

- (e) Now assume that the reverse occurs, that is, the output of the comparator has railed to the bottom rail. Repeat part (d) with this value of v_3 .

- (f) What is v_3 as a function of time? Draw a graph of v_3 and v_2 . Since the graph is periodic, find its period and frequency.

- (g) Suppose that we changed the value of C_1 to be $2\mu\text{F}$? What is the new period? Suppose that we change R_5 to be $2\text{k}\Omega$. What is the new period? What if we change R_5 to be 0Ω ? Will this circuit still operate?

EECS 16A Designing Information Devices and Systems I

Discussion 13A

Reference: Inner products

Let $\vec{x}$, $\vec{y}$, and $\vec{z}$ be vectors in real vector space $\mathbb{V}$. A mapping $\langle \cdot, \cdot \rangle$ is said to be an inner product on $\mathbb{V}$ if it satisfies the following three properties:

- (a) Symmetry: $\langle \vec{x}, \vec{y} \rangle = \langle \vec{y}, \vec{x} \rangle$.
- (b) Linearity: $\langle \vec{x}, \vec{y} + \vec{z} \rangle = \langle \vec{x}, \vec{y} \rangle + \langle \vec{x}, \vec{z} \rangle$ and $\langle c\vec{x}, \vec{y} \rangle = c\langle \vec{x}, \vec{y} \rangle$.
- (c) Non-negativeness: $\langle \vec{x}, \vec{x} \rangle \geq 0$, with equality if and only if $\vec{x} = \vec{0}$.

$$\begin{aligned}
 \langle \vec{x}, \vec{y} \rangle &\equiv \vec{x} \cdot \vec{y} \equiv \vec{x}^T \vec{y} \\
 &= \begin{bmatrix} x_1 & x_2 & \cdots & x_n \end{bmatrix} \begin{bmatrix} y_1 \\ y_2 \\ \vdots \\ y_n \end{bmatrix} \\
 &= x_1 y_1 + x_2 y_2 + \cdots + x_n y_n \\
 &= \sum_{i=1}^n x_i y_i.
 \end{aligned}$$

We define the norm of $\vec{x} = [x_1, x_2, \dots, x_n]^T$ as $\|\vec{x}\| = \sqrt{x_1^2 + x_2^2 + \dots + x_n^2}$.

1. Geometric Interpretation of the Inner Product

In this problem, we will explore the geometric interpretation of the Euclidean inner product, restricting ourselves to vectors in $\mathbb{R}^2$.

- (a) For each of the following cases, pick two vectors that satisfy the condition and find the inner product.

- i. Parallel:

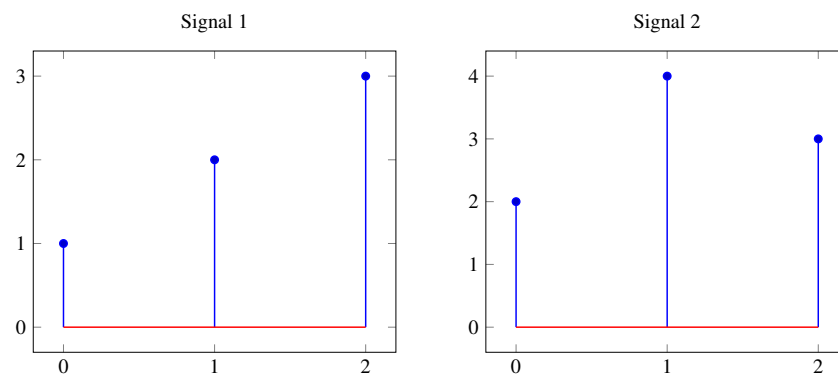
- ii. Anti-parallel:

iii. Perpendicular:

- (b) Now, derive a formula for the inner product of two vectors in terms of their magnitudes and the angle between them.

2. Correlation

We are given the following two signals, $\vec{s}_1[n]$ and $\vec{s}_2[n]$ respectively.



Find the cross correlations, $\text{corr}_{\vec{s}_1}(\vec{s}_2)$ and $\text{corr}_{\vec{s}_2}(\vec{s}_1)$ for signals $\vec{s}_1[n]$ and $\vec{s}_2[n]$. Recall

$$\text{corr}_{\vec{x}}(\vec{y})[k] = \sum_{i=-\infty}^{\infty} x[i]y[i-k].$$

	$\text{corr}_{\vec{s}_1}(\vec{s}_2)[k]$						
$\vec{s}_1$	0	0	1	2	3	0	0
$\vec{s}_2[n+2]$							
$\langle \vec{s}_1, \vec{s}_2[n+2] \rangle$	+	+	+	+	+	+	=

	$\text{corr}_{\vec{s}_2}(\vec{s}_1)[k]$						
$\vec{s}_1$	0	0	1	2	3	0	0
$\vec{s}_2[n+1]$							
$\langle \vec{s}_1, \vec{s}_2[n+1] \rangle$	+	+	+	+	+	+	=

$\vec{s}_1$	0	0	1	2	3	0	0
$\vec{s}_2[n]$							
$\langle \vec{s}_1, \vec{s}_2[n] \rangle$	+	+	+	+	+	+	=

$\vec{s}_1$	0	0	1	2	3	0	0
$\vec{s}_2[n-1]$							
$\langle \vec{s}_1, \vec{s}_2[n-1] \rangle$	+	+	+	+	+	+	=

$\vec{s}_1$	0	0	1	2	3	0	0
$\vec{s}_2[n-2]$							
$\langle \vec{s}_1, \vec{s}_2[n-2] \rangle$	+	+	+	+	+	+	=

	$\text{corr}_{\vec{s}_2}(\vec{s}_1)[k]$						
$\vec{s}_2$	0	0	2	4	3	0	0
$\vec{s}_1[n+2]$							
$\langle \vec{s}_2, \vec{s}_1[n+2] \rangle$	+	+	+	+	+	+	=

$\vec{s}_2$	0	0	2	4	3	0	0
$\vec{s}_1[n+1]$							
$\langle \vec{s}_2, \vec{s}_1[n+1] \rangle$	+	+	+	+	+	+	=

$\vec{s}_2$	0	0	2	4	3	0	0
$\vec{s}_1[n]$							
$\langle \vec{s}_2, \vec{s}_1[n] \rangle$	+	+	+	+	+	+	=

$\vec{s}_2$	0	0	2	4	3	0	0
$\vec{s}_1[n-1]$							
$\langle \vec{s}_2, \vec{s}_1[n-1] \rangle$	+	+	+	+	+	+	=

$\vec{s}_2$	0	0	2	4	3	0	0
$\vec{s}_1[n-2]$							
$\langle \vec{s}_2, \vec{s}_1[n-2] \rangle$	+	+	+	+	+	+	=

3. Inner Product Properties

Prove the following properties of inner products for any vectors in $\mathbb{R}^2$, assuming we are working with the Euclidean inner product and norm.

(a) Symmetry: $\langle \vec{x}, \vec{y} \rangle = \langle \vec{y}, \vec{x} \rangle$.

(b) Linearity: $\langle \alpha \vec{x} + \beta \vec{y}, \vec{z} \rangle = \alpha \langle \vec{x}, \vec{z} \rangle + \beta \langle \vec{y}, \vec{z} \rangle$.

EECS 16A Designing Information Devices and Systems I

Discussion 14A

1. Search and Rescue Dogs

Berkeley's Puppy Pound needs your help! While Mr. Muffin was being walked, the volunteer let go of his leash and he is now running wild in the streets of Berkeley (which are quite dangerous)! Thankfully, all of the puppies at the pound have a collar that sends a bluetooth signal to receiver towers, which are spread throughout the streets (pictured below). If the puppy/collar is within range of the receiver tower, the collar will send the tower a message: the distance of the collar to the tower. Each cell tower has a range of 5 city blocks. Can you help the pound locate their lost puppy?

Note: A city block is defined as the middle of an intersection to the middle of an adjacent intersection (scale provided on map.) Mr. Muffin is constrained to running wild in the streets, meaning he won't be found in any buildings. If your TA asks 'Where is Mr. Muffin?' it is sufficient to answer with 'this intersection' or 'between these two intersections.'

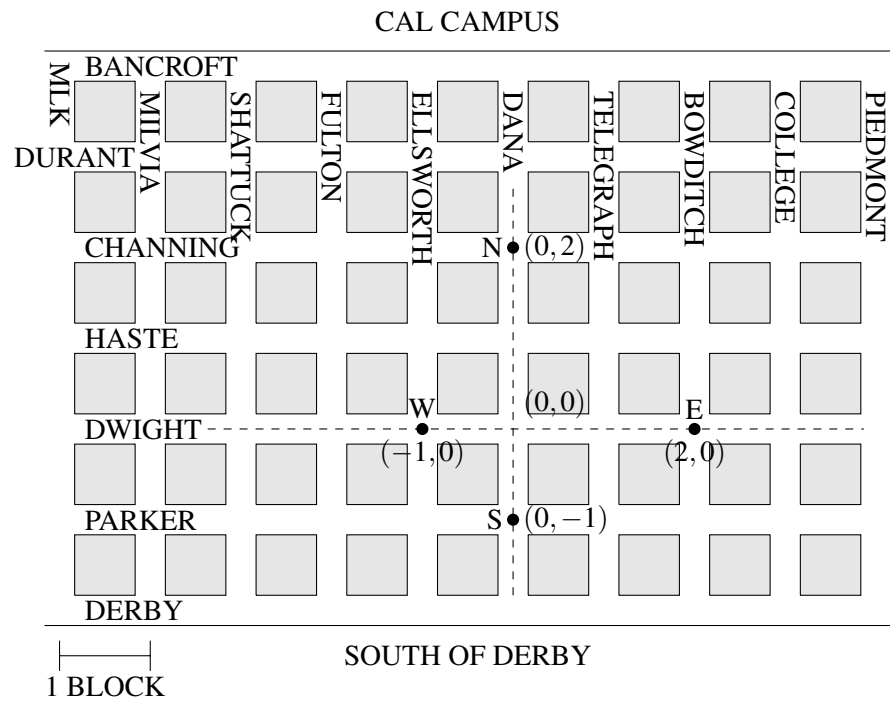


(a) You check the logs of the cell towers, and they have received the following messages:

Sensor	Distance
N	5
W	$\sqrt{20}$
E	$\sqrt{5}$
S	$\sqrt{10}$

On the map provided, identify where Mr. Muffin is!

¹http://www.pupsmile.com/wp-content/uploads/2012/11/running_happy_dog-1024x684.jpeg



- (b) Can you set this up as a system of equations? Is it linear? If it's not linear, can you think of a way to make it linear? Now, how do you set this up in matrix form?

Hint: Set (0,0) to be Dwight and Dana.

Hint 2: Distance = $\sqrt{(x_1 - x_2)^2 + (y_1 - y_2)^2}$.

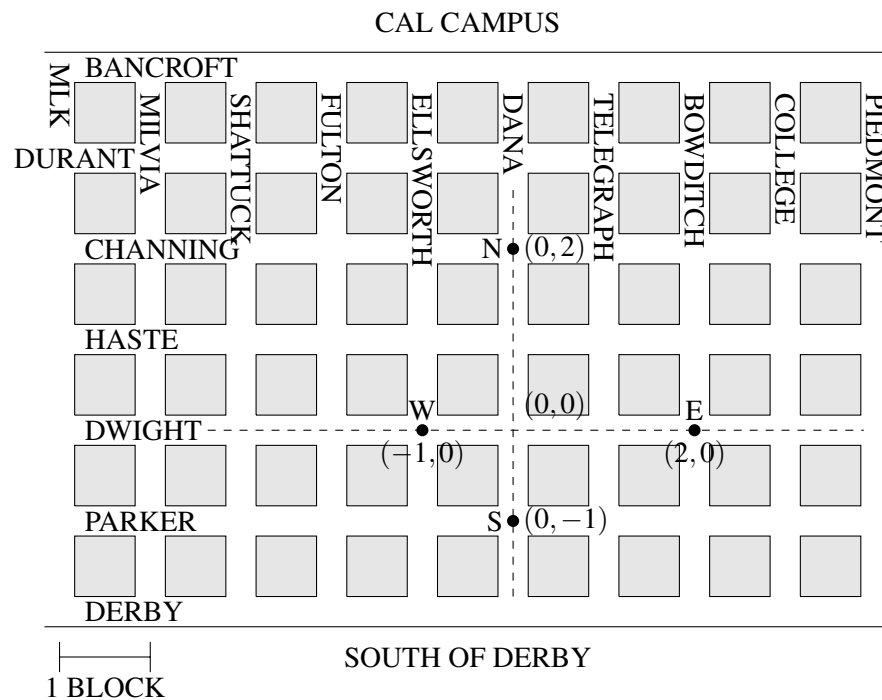
Hint 3: You don't need all 4 equations. You have two unknowns, x and y . You know from lecture that you need three circles to uniquely find a point. How can you use the third circle/equation to get two equations and two unknowns?

Note: Remember to check for consistency for all nonlinear equations after finding the coordinates.

- (c) Suppose Mr. Muffin is moving fast, and by the time you get to destination in part (a) he's already run off! You check the logs of the cell towers again, and see the following updated messages:

Sensor	Distance
N	5
W	$\sqrt{20}$
E	Out of Range
S	Out of Range

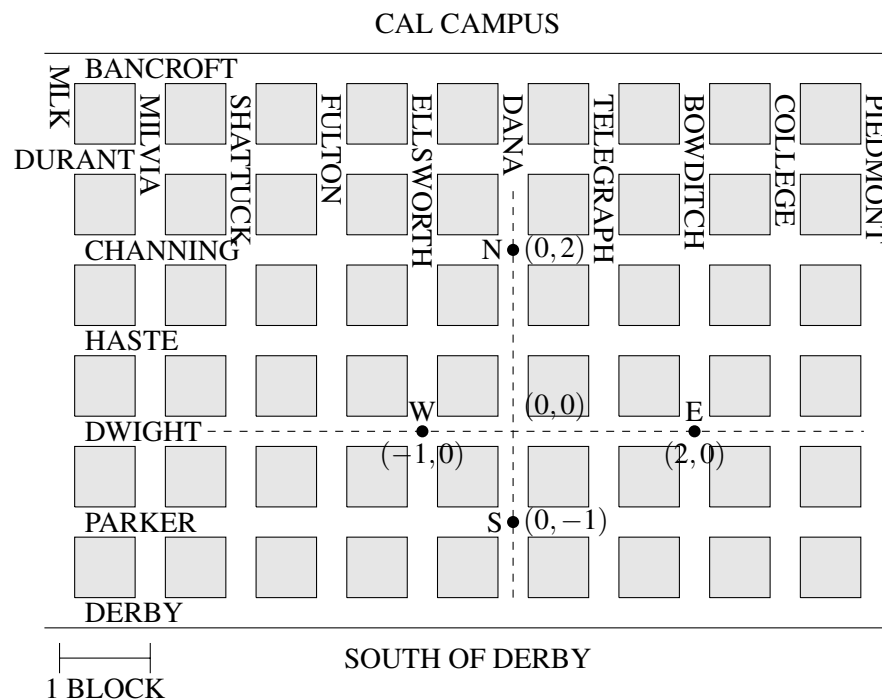
Can you find Mr. Muffin? (*Note: Each cell tower has a range of 5 city blocks.*)



- (d) Mr. Muffin is a very mischievous puppy, and while playing and running around he damaged his collar. The transmitter on his collar will still send a signal to the receiver towers, but the distance sensor has noise. You check the logs of the cell towers, and they have received the following messages:

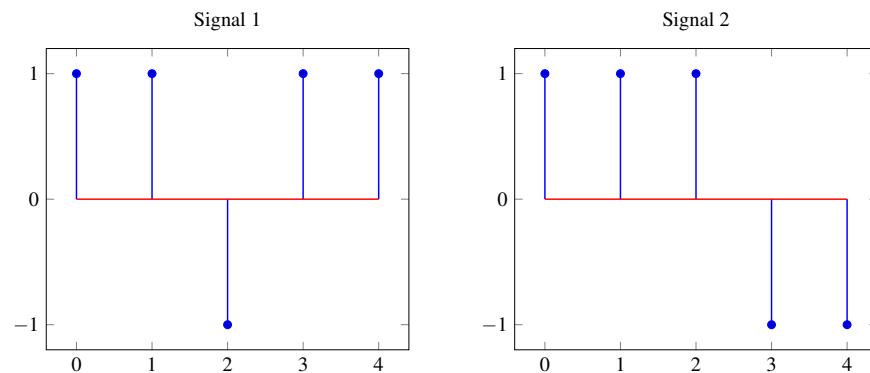
Sensor	Distance
N	4.5 ± 0.5
W	Out of Range
E	4.5 ± 0.5
S	Out of Range

On the map provided, identify where Mr. Muffin is! Can you find exactly where he is? (*Note: Each cell tower has a range of 5 city blocks.*)

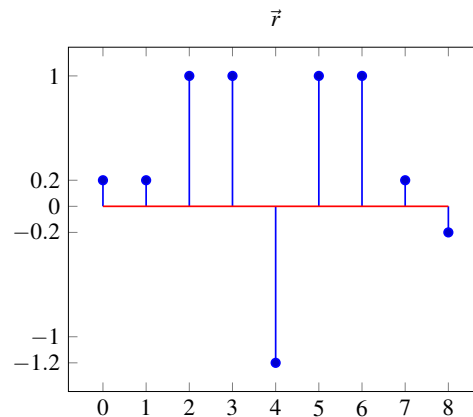


2. Identifying satellites and their delays

We are given the following two signals, $\vec{s}_1$ and $\vec{s}_2$ respectively, that are signatures for two satellites.

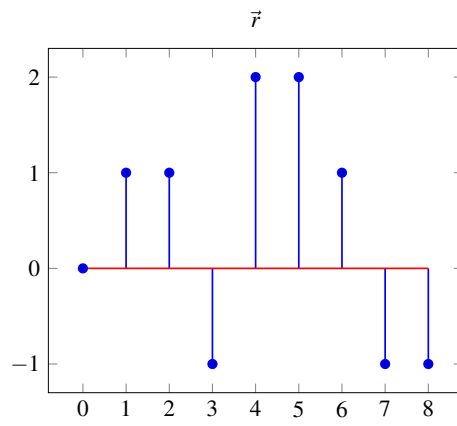


- (a) Your cellphone antenna receives the following signal $r[n]$. You know that there may be some noise present in $r[n]$ in addition to the transmission from the satellite.



Which satellites are transmitting? What is the delay between the satellite and your cellphone? Use cross-correlation to justify your answer. You can use iPython to compute the cross-correlation.

- (b) Now your cellphone receives a new signal $r[n]$ as below. Which satellites are transmitting and what is the delay between each satellite and your cellphone?



3. From Inner Products To Projections

Given that $\langle \vec{x}, \vec{y} \rangle$ is a measure of similarity between two vectors, let's try to use this to find how much of one vector $\vec{y}$ is in the direction of another vector $\vec{x}$.

- (a) Let's start with $\langle \vec{x}, \vec{y} \rangle$. We want a quantity that is independent of the norm of $\vec{x}$, $\|\vec{x}\|$. Is $\langle \vec{x}, \vec{y} \rangle$ independent of the norm? Consider $\langle \vec{x}, \vec{y} \rangle$ for the examples below.

$$\vec{x} = \begin{bmatrix} 1 \\ 1 \end{bmatrix}, \vec{y} = \begin{bmatrix} 2 \\ 1 \end{bmatrix} \text{ and } \vec{x} = \begin{bmatrix} 2 \\ 2 \end{bmatrix}, \vec{y} = \begin{bmatrix} 2 \\ 1 \end{bmatrix}$$

- (b) Suppose we divide $\langle \vec{x}, \vec{y} \rangle$ by the norm of $\vec{x}$, $\|\vec{x}\|$, to get $\frac{\langle \vec{x}, \vec{y} \rangle}{\|\vec{x}\|}$. Is this new quantity independent of the norm of $\vec{x}$? Test it on the examples above.

- (c) We now have a scalar quantity that represents how much of $\vec{y}$ is in the direction of $\vec{x}$. Now let's look for a vector $\vec{z}$ that has a norm of $\frac{\langle \vec{x}, \vec{y} \rangle}{\|\vec{x}\|}$ and points in the same direction as $\vec{x}$. The vector $\vec{z}$ should be written in terms of $\vec{x}$ and $\vec{y}$.

- (d) Consider the quantity $\frac{\langle \vec{x}, \vec{y} \rangle}{\|\vec{x}\| \|\vec{y}\|}$. What is the maximum this quantity could be? When does this occur? What is the minimum this quantity could be? When does this occur?

Hint: Use the Cauchy-Schwartz inequality.

- (e) We define the angle between two vectors as $\cos(\theta) = \frac{\langle \vec{x}, \vec{y} \rangle}{\|\vec{x}\| \|\vec{y}\|}$. When do two vectors have an angle of 90° between them? When do they have an angle of 0° ? When do they have an angle of 180° ?

EECS 16A Designing Information Devices and Systems I

Discussion 14B

1. Least Squares with Orthogonal Columns

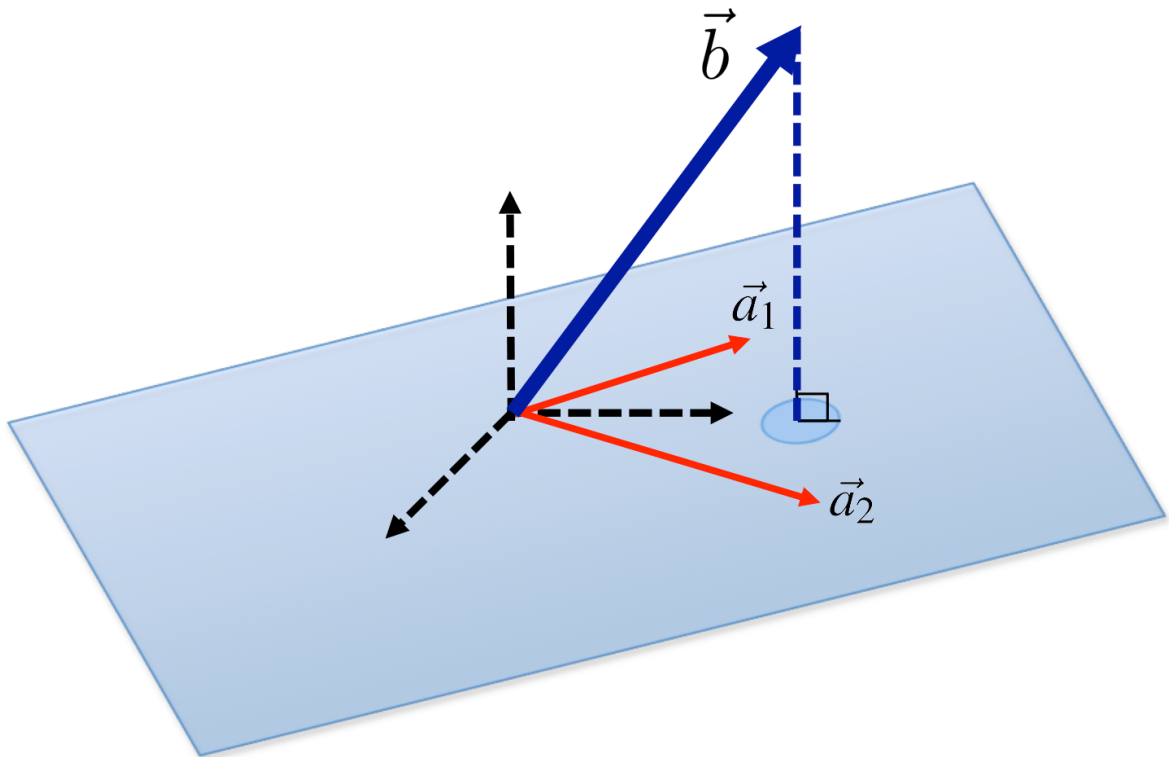
(a) Consider a least squares problem of the form

$$\min_{\vec{x}} \left\| \vec{b} - \mathbf{A}\vec{x} \right\|^2 = \min_{\vec{x}} \left\| \mathbf{A}\vec{x} - \vec{b} \right\|^2 = \min_{\vec{x}} \left\| \begin{bmatrix} b_1 \\ b_2 \\ b_3 \end{bmatrix} - \begin{bmatrix} | & | \\ \vec{a}_1 & \vec{a}_2 \\ | & | \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} \right\|^2$$

Let the solution be $\vec{\hat{x}} = \begin{bmatrix} \hat{x}_1 \\ \hat{x}_2 \end{bmatrix}$.

Label the following elements in the diagram below.

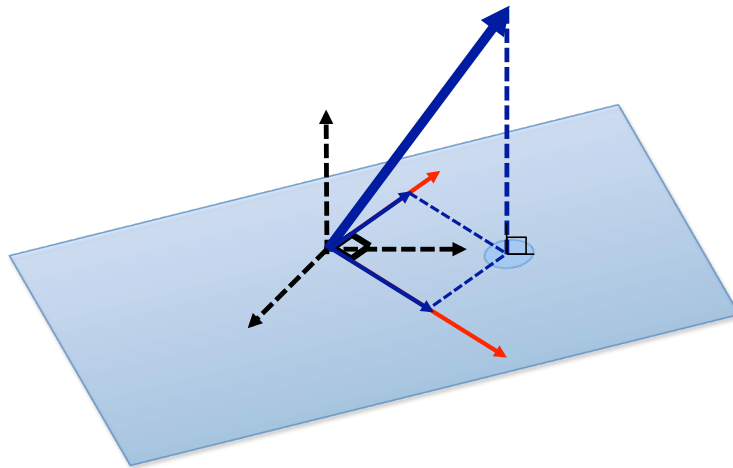
$\text{span}\{\vec{a}_1, \vec{a}_2\}$, $\vec{e} = \vec{b} - \mathbf{A}\vec{\hat{x}}$, $\mathbf{A}\vec{\hat{x}}$, $\vec{a}_1\hat{x}_1, \vec{a}_2\hat{x}_2$, $\text{colspace}(\mathbf{A})$



- (b) We now consider the special case of least squares where the columns of $\mathbf{A}$ are orthogonal (illustrated in the figure below). Given that $\vec{x} = (\mathbf{A}^T \mathbf{A})^{-1} \mathbf{A}^T \vec{b}$ and $\mathbf{A} \vec{x} = \text{proj}_{\mathbf{A}}(\vec{b}) = \hat{x}_1 \vec{a}_1 + \hat{x}_2 \vec{a}_2$, show that

$$\text{proj}_{\vec{a}_1}(\vec{b}) = \hat{x}_1 \vec{a}_1$$

$$\text{proj}_{\vec{a}_2}(\vec{b}) = \hat{x}_2 \vec{a}_2$$



- (c) Compute the least squares solution to

$$\min_{\vec{x}} \left\| \begin{bmatrix} 1 \\ 2 \\ 3 \end{bmatrix} - \begin{bmatrix} 1 & 0 \\ 0 & 0 \\ 0 & 1 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} \right\|^2.$$

2. Polynomial Fitting

Even though least squares can only be applied to linear systems, it turns out that it can also solve problems with decidedly nonlinear elements. In lecture, you will see an example of fitting to an ellipse. Here, we will fit to a nonlinear polynomial.

Say we know that y is a quartic polynomial in x . This means that we know that y and x are related as follows:

$$y = a_0 + a_1x + a_2x^2 + a_3x^3 + a_4x^4$$

We're also given the following observations, and our goal is to figure out the relationship between x and y :

x	y
0.0	24.0
0.5	6.61
1.0	0.0
1.5	-0.95
2.0	0.07
2.5	0.73
3.0	-0.12
3.5	-0.83
4.0	-0.04
4.5	6.42

(a) What are the unknowns in this question?

(b) Can you write an equation corresponding to the first observation (x_0, y_0) , in terms of a_0, a_1, a_2, a_3 , and a_4 ? What does this equation look like? Is it linear in the unknowns?

(c) Now, write a system of equations in terms of a_0 , a_1 , a_2 , a_3 , and a_4 using *all of the observations*.

(d) Finally, solve for a_0 , a_1 , a_2 , a_3 , and a_4 using IPython or any method you like. You have now found the quartic polynomial that best fits the data!

(e) What if we didn't know the degree of the polynomial? Use the IPython Notebook to explore what happens when we choose a polynomial degree other 4 and explain what you see.

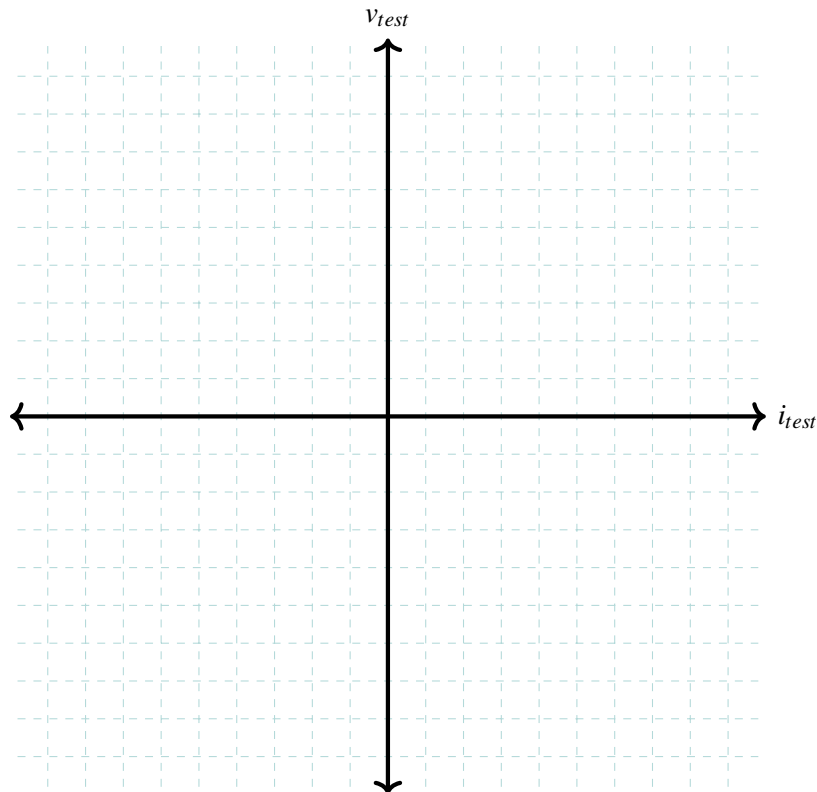
- (f) **OPTIONAL:** Play around with what happens when you add more noise to the data or if you decide to drop data points on the IPython Notebook. Additionally, explore what you see when you change the degree of the polynomial alongside these factors.

3. Ohm's Law With Noise (Optional)

We are trying to measure the resistance of a black box. We apply various i_{test} currents and measure the output voltage v_{test} . Sometimes, we are quite fortunate to get nice numbers. Oftentimes, our measurement tools are a little bit noisy, and the values we get out of them are not accurate. However, if the noise is completely random, then the effect of it can be averaged out over many samples. So we repeat our test many times:

Test	i_{test} (mA)	v_{test} (V)
1	10	21
2	3	7
3	-1	-2
4	5	8
5	-8	-15
6	-5	-11

(a) Plot the measured voltage as a function of the current.



- (b) Suppose we stack the currents and voltages to get $\vec{I} = \begin{bmatrix} 10 \\ 3 \\ -1 \\ 5 \\ -8 \\ -5 \end{bmatrix}$ and $\vec{V} = \begin{bmatrix} 21 \\ 7 \\ -2 \\ 8 \\ -15 \\ -11 \end{bmatrix}$. Is there a unique solution for R ? What conditions must $\vec{I}$ and $\vec{V}$ satisfy in order for us to solve for R uniquely?

- (c) Ideally, we would like to find R such that $\vec{V} = \vec{I}R$. If we cannot do this, we'd like to find a value of R that is the *best* solution possible, in the sense that $\vec{I}R$ is as “close” to $\vec{V}$ as possible. We are defining the sum of squared errors as a **cost function**. In this case the cost function for any value of R quantifies the difference between each component of $\vec{V}$ (i.e. v_j) and each component of $\vec{I}R$ (i.e. i_jR) and sum up the squares of these “differences” as follows:

$$\text{cost}(R) = \sum_{j=1}^6 (v_j - i_jR)^2$$

Do you think this is a good cost function? Why or why not?

- (d) Show that you can also express the above cost function in vector form, that is,

$$\text{cost}(R) = \langle (\vec{V} - \vec{I}R), (\vec{V} - \vec{I}R) \rangle$$

Hint: $\langle \vec{a}, \vec{b} \rangle = \vec{a}^T \vec{b} = \sum_i a_i b_i$

- (e) Find $\hat{R}$, which is defined as the optimal value of R that minimizes $\text{cost}(R)$.

Hint: Use calculus. The optimal $\hat{R}$ makes $\frac{d\text{cost}(\hat{R})}{dR} = 0$

- (f) On your original IV plot, also plot the line $v_{test} = \hat{R}i_{test}$. Can you visually see why this line “fits” the data well? How well would we have done if we had guessed $R = 3\text{ k}\Omega$? What about $R = 1\text{ k}\Omega$? Calculate the cost functions for each of these choices of R to validate your answer.

- (g) Now, suppose that we add a new data point: $i_7 = 2 \text{ mA}$, $v_7 = 4 \text{ V}$. Will $\hat{R}$ increase, decrease, or remain the same? Why? What does that say about the line $v_{test} = \hat{R}i_{test}$?

4. Orthonormal Matrices and Projections (Optional)

An orthonormal matrix, $\mathbf{A}$, is a matrix whose columns, $\vec{a}_i$, are:

- Orthogonal (ie. $\langle \vec{a}_i, \vec{a}_j \rangle = 0$ when $i \neq j$)
- Normalized (ie. vectors with length equal to 1, $\|\vec{a}_i\| = 1$). This implies that $\|\vec{a}_i\|^2 = \langle \vec{a}_i, \vec{a}_i \rangle = 1$.

(a) Suppose that the matrix $\mathbf{A} \in \mathbb{R}^{N \times M}$ has linearly independent columns. The vector $\vec{y}$ in $\mathbb{R}^N$ is not in the subspace spanned by the columns of $\mathbf{A}$. What is the projection of $\vec{y}$ onto the subspace spanned by the columns of $\mathbf{A}$?

(b) Show if $\mathbf{A} \in \mathbb{R}^{N \times N}$ is an orthonormal matrix then the columns, $\vec{a}_i$, form a basis for $\mathbb{R}^N$.

(c) When $\mathbf{A} \in \mathbb{R}^{N \times M}$ and $N \geq M$ (i.e. tall matrices), show that if the matrix is orthonormal, then $\mathbf{A}^T \mathbf{A} = \mathbf{I}_{M \times M}$.

- (d) Again, suppose $\mathbf{A} \in \mathbb{R}^{N \times M}$ where $N \geq M$ is an orthonormal matrix. Show that the projection of $\vec{y}$ onto the subspace spanned by the columns of $\mathbf{A}$ is now $\mathbf{A}\mathbf{A}^T\vec{y}$.

- (e) Given $\mathbf{A} \in \mathbb{R}^{N \times M} = \begin{bmatrix} 0 & 0 & \frac{\sqrt{2}}{2} \\ 0 & 0 & \frac{\sqrt{2}}{2} \\ 0 & 1 & 0 \\ 1 & 0 & 0 \end{bmatrix}$ and the columns of $\mathbf{A}$ are orthonormal, find the least squares solution to $\mathbf{A}\hat{\vec{x}} = \vec{y}$ where $\vec{y} = [5 \quad 12 \quad 7 \quad 8]^T$.